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Before the Vote: The Temporal Structure and Workforce Composition of Political
Information Supply on Wikipedia on U.S. House Candidate Pages, 2008–2024

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March 2026

Abstract

As a high-visibility, low-cost reference good, Wikipedia candidate biography pages are a prominent source of political information for voters, yet little is known about when and by whom this content is produced over the election cycle. This paper analyzes the temporal structure and workforce composition of editing on U.S. House candidate pages across nine election cycles (2008–2024), using complete revision histories from Wikipedia’s public API. An event-study regression finds a modest, statistically significant pre-election ramp in editing activity during the post-primary, pre-general-election window. A second design exploiting cross-state variation in primary dates finds no evidence that editing tracks state-specific primary calendars once nationally synchronized forces are controlled for. Descriptive analysis of editor portfolios reveals an increasingly geographically expansive editing workforce, with the highest-volume contributors editing candidates of both parties across many states in a pattern consistent with informational maintenance rather than partisan advocacy. The pre-election ramp does not much alter the composition of editing produced. These findings extend the nationalization thesis from voter behavior and media coverage to the supply side of a major reference platform.

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1. Introduction

As one of the most popular websites in the US (Semrush 2025), the English site of Wikipedia generated roughly 85 billion page views in 2025 alone (Wikimedia Pageviews API 2026a). Being a high-visibility, low-cost reference good, Wikipedia offers a quick, seemingly neutral overview of a candidate, especially for readers who are not already deeply engaged, thus occupying an unusual position in the U.S. political information environment. In the months surrounding a national election, candidate and officeholder pages attract a vast number of queries from voters, campaign researchers (Yasseri 2015) and the systems that aggregate and synthesize political information at scale. A growing empirical literature has used Wikipedia pageview data as a behavioral proxy for information seeking, documenting clear demand-side spikes in candidate-page traffic during parliamentary elections and controversies (Lupacheva 2025). Moreover, online Wikipedia traffic patterns have been shown to track electoral salience in ways that add predictive signal to forecasting models (Yasseri and Bright 2016; Smith and Gustafson 2017). This body of work establishes that Wikipedia is consequential for how political information reaches the public. However, less focus has been placed on the supply side of political and electoral information on Wikipedia. It remains systematically undercharacterized how such information is continuously written, revised, and contested by a community of volunteer editors over the election cycle.

This paper focuses on the production side of Wikipedia as a political information system. In contrast to professionally produced mass media and platform-mediated

communication infrastructures, Wikipedia is a large-scale, rule-governed peer production system in which volunteer contributors coordinate through procedures, norms, and governance institutions. Candidate biographies are written under formal content constraints, such as neutrality and verifiability, and are continuously policed through community quality-control practices. Yet there is no central editor assigning beats or certifying a final version of information for dissemination. What millions of readers encounter at any moment is the current state of an ongoing, decentralized process. What a voter reads about a candidate during an election campaign reflects the accumulated edits made by some set of contributors at earlier moments, under particular incentives, and at particular distances from the electoral calendar. On the other hand, the Wikipedia articles are not terminal either; as they become embedded in search-engine knowledge panels and increasingly used as training data for large language models, the political content produced by editors flows outward into the broader information ecosystem. Thus, the production side per se is politically consequential, and characterizing this production process is a prerequisite for understanding Wikipedia’s role as political information infrastructure.

The editing behavior that constitutes production can arise from at least three analytically distinct, but non-mutually exclusive, mechanisms. The first is *reactive supply*: editors respond to shocks in the surrounding information environment by updating articles to reflect a changing record; such shocks include more news coverage, more allegations that must be sourced or removed, and more disputes that must be adjudicated (Keegan, Gergle, & Contractor 2013). Under this mechanism, Wikipedia acts as a downstream integrator of other information institutions, translating pub-

licly available reporting into encyclopedic prose. The second is *electoral engagement*: independent of any particular scandal or policy announcement, elections themselves can shift attention and effort toward political pages; this is consistent with classic ideas about attention cycles, civic mobilization, and the allocation of volunteer labor (Downs 1972). Under this mechanism, the community’s own heightened attentiveness as high-salience moments approach is reflected in the timing and volume of new revisions. The third is *strategic information production*: because candidate pages are highly visible and widely reused, interested actors may attempt to shape what is emphasized, how it is framed, and which sources are treated as authoritative (Kalla & Aronow 2015). Importantly, this does not imply unconstrained success; influence attempts must pass through Wikipedia’s procedural filters and are often met by counter-editing or page-protection enforcement.

These diverse mechanisms in Wikipedia’s political information production suggest that political information supply on Wikipedia is not temporally uniform, though it may be hard to exactly pinpoint which mechanism(s) dominate. This paper, then, starts from some basic but unanswered descriptive questions: *how does the volume and concentration of editing on candidate biographies usually evolve over each election cycle, and how has that changed over recent years?* Specifically, I focus on the pre-election editing behavior since it is the primary opportunity for candidate information to be made accurate, current and complete as the most high-stakes readership (voters) arrives.

Despite these high practical and normative stakes of Wikipedia in electoral information supply, we have surprisingly little systematic evidence on the basic objects:

when political biographies on Wikipedia are produced over the election cycle, how concentrated that work is, and whether it is organized around local electoral calendars or a common national rhythm. Providing that characterization is the core contribution of this paper, and it sets the context for interpreting any claims about Wikipedia’s role in democratic accountability.

This paper analyzes the temporal structure of information production on English Wikipedia for U.S. House candidate biography pages across nine election cycles (2008–2024), using complete revision histories from Wikipedia’s public API. Two complementary empirical designs are employed. A within-cycle event-study regression establishes whether editing volume rises systematically in the post-primary, pre-general window, net of stable page differences and secular trends in Wikipedia-wide activity. A design exploiting cross-state variation in primary dates then asks whether any observed buildup tracks state-specific electoral calendars or is instead driven by national synchronization, placing the production-side question in direct dialogue with the American politics literature on the nationalization of the information environment (Hopkins 2018; Moskowitz 2021). The event study finds a modest but statistically significant pre-election ramp that characterizes both the strictly post-primary window and the conventional post-Labor Day campaign sprint (Fowler, Franz, and Ridout 2021; Erikson and Wlezien 2012). The state-primary timing analysis finds little evidence that production is organized around state-specific primary calendars once nationally synchronized forces are controlled for. Complementary analysis of editor portfolios shows an increasingly geographically expansive editing workforce from 2008 to 2024, with the highest-volume contributors tending to edit candidates of both par-

ties across many states. This suggests the growth of maintenance-oriented generalists rather than partisan actors. A direct comparison of editing composition between baseline and near-general-election windows further reveals that the pre-election ramp does not alter the cross-party composition of the editing workforce, though editors in the pre-election window are slightly more geographically concentrated in their top state. The paper proceeds as follows. Section 2 develops the empirical literature context. Sections 3 and 4 describe the data and methods. Sections 5 and 6 present regression and descriptive results. Section 7 discusses implications and limitations.

2. Literature Review

A large empirical literature has used Wikipedia pageviews as a behavioral proxy for election-season information seeking, documenting election-time spikes in attention and showing that Wikipedia usage can add predictive signal to election forecasting models even when it does not independently explain outcomes (Yasseri & Bright 2014, 2016; Smith & Gustafson 2017; Agarwal et al. 2020; Oswald & Ohrenhofer 2022; Oswald 2025). This prediction-oriented tradition shows that Wikipedia matters during campaigns, but it primarily concerns consumption (views) rather than production.

A separate body of research that does focus on content production on Wikipedia takes the perspective of large-scale peer production and collective intelligence systems, where revision histories provide traces of how information is assembled, contested, and maintained. Such work on Wikipedia highlights both the “wisdom of the crowd” and the “power of the few,” showing that openness to many contributors can coexist with highly unequal distributions of labor and influence (Arazy, Morgan, &

Patterson 2006; Kittur et al. 2007). Subsequent research has emphasized that editors occupy differentiated roles and that participation is shaped by coordination, quality control, and gatekeeping processes; for example, newcomer experiences and reversion dynamics can affect who continues to contribute during bursts of attention (Halfaker, Kittur, & Riedl 2011; Halfaker et al. 2013; Müller-Birn et al. 2015; Piscopo & Simperl 2018). More broadly, research using interaction networks and signed ties among editors has shown how conflict, coordination, and team structure relate to information production and, in some cases, article quality (Adler & de Alfaro 2007; Flöck & Acosta 2014; Lerner & Lomi 2017, 2019). My project takes a similar epistemological foundation to these works by treating pre-election editing as a structured rather than purely idiosyncratic process, with a systematic component potentially related to the electoral environment and stochastic editor-level variations.

Within political pages specifically, prior scholarship argues that *elections and strategic incentives* can be visible in revision patterns. Biographical pages can function as a *marketplace* for political information; they are highly discoverable, frequently consulted by less-engaged voters, and therefore attractive sites for informational influence during campaigns (Göbel & Munzert 2017). Experimental evidence also suggests that incentives are high-stakes for active politicians. Specifically, negative information has been found to be removed more readily than positive information from the biographies of current U.S. senators, and this pattern appears weaker for retired or deceased officials (Kalla & Aronow 2015). Other work points to persistent tensions over political representation and identity, including patterns of polarization and conflict in political-page editing (Umarova & Mustafaraj 2019), and to the idea that

partisan effects may be mitigated in some settings by Wikipedia’s editing ecology (Pradel 2020). These studies suggest that election periods are likely to be moments of heightened activity and can potentially alter participation dynamics on candidate pages.

Much of such political-Wikipedia literature either focuses on particular institutional contexts (e.g., specific countries or elite subsets) or is oriented toward outcomes like content quality and bias. For example, work on UK Members of Parliament documents election-time surges in engagement and suggests that editor activity can cluster by party in their editing portfolios, with a small fraction of editors accounting for a large fraction of edits; importantly, it also emphasizes that such clustering need not imply partisan bias if it reflects asymmetric media coverage and information availability (Agarwal et al. 2020). Thus far, no paper has focused on empirically investigating *pre-election temporal editing dynamics*. To fill such a gap, this study uses U.S. House candidates over multiple cycles as the setting.

The choice of U.S. House races as the empirical setting connects this descriptive project to a broader question in the American politics literature, i.e., whether the information environment surrounding down-ballot elections is becoming more nationally organized. A growing body of work argues that electoral behavior and representation are increasingly much more shaped by national partisan forces than district- or state-specific considerations (Hopkins 2018; Jacobson 2015; Shor & McCarty 2011), and that this shift is partly an informational phenomenon. When local news coverage declines, voters evaluate candidates through national lenses, and down-ballot contests become more tightly linked to national-level dynamics (Moskowitz 2021). If

the broader information environment surrounding House elections is indeed becoming more national, one might expect this to be visible in how political information is produced on Wikipedia as well, including editing activity primarily organized around a common national electoral rhythm rather than state-specific primary calendars, and an editing workforce whose portfolios span many states.

In this paper, I use some empirical designs to measure exactly these production-side patterns. The state-primary timing analysis directly tests whether editing follows local electoral calendars or is instead nationally synchronized, by exploiting cross-state variation in primary dates within shared calendar weeks. The descriptive analysis of editor portfolios measures the geographic and partisan scope of individual editors' contributions across election cycles. These two approaches taken together characterize the temporal and spatial organization of candidate-page editing. Such organization is observable in Wikipedia revision histories and has clear analogs in the nationalization literature's account of how political attention is structured.

To be precise about the inferential boundary, this paper documents patterns in *information production*, not in electoral behavior or voter decision-making. Editing that is nationally timed and geographically expansive is consistent with a more nationalized information environment, but it is not uniquely diagnostic of it; the same patterns could arise from platform-specific factors internal to Wikipedia (e.g., the maturation of cross-state editing tools or WikiProject coordination). What I attempt to establish is whether the temporal and geographic organization of Wikipedia editing on House-candidate pages *looks increasingly national*, and whether that characterization holds across multiple election cycles. That is a narrower claim than electoral

nationalization, but it is a necessary empirical input for any account of Wikipedia’s role in the political information ecosystem.

3. Hypotheses

3.1. Regression

H1 (National election-season ramp). *Editing on House candidate pages increases in the post-primary pre-general window, after controlling for page-cycle differences and secular year-to-year trends in Wikipedia activity.*

Formally,

$$NI = \frac{1}{8} \sum_{k=-8}^{-1} \hat{\beta}_k^{(G)} > 0,$$

where $\hat{\beta}_k^{(G)}$ is estimated from the within-cycle event study regression (Section 4.2 Equation 1). The window $k \in \{-8, \dots, -1\}$ is chosen to fall after all major state primaries have concluded and post-primary page-updating bursts have largely dissipated. This hypothesis is falsified if NI is indistinguishable from zero.

H2 (State-primary timing signal). *After controlling for national synchronization and stable page-cycle differences, editing increases on a political candidate’s page as it approaches the state primary that the candidate runs for.*

Formally, for at least some $k \in \{-18, \dots, -1\}$, $\beta_k > 0$, and the near-primary average $\bar{\beta}_K = \frac{1}{|K|} \sum_{k \in K} \beta_k$ is positive for a near window K . The pooled β_k estimates from the baseline Model 1 (Section 4.2 Equation 2) identify the average state-primary timing effect across all cycles; testing $H_0 : \bar{\beta}_K^{(M1)} = 0$ against the one-sided alternative directly addresses whether any residual state-primary timing signal exists on average. Under H2, the expected pattern is a systematic upward shift in $\hat{\beta}_k$ as k approaches -1 , with $\bar{\beta}_K^{(M1)} > 0$. This hypothesis is falsified if $\hat{\beta}_k$ fluctuates near zero across k and if $\bar{\beta}_K^{(M1)}$ is not distinguishable from zero.

H3 (Attenuation of state-primary timing). *House politics becomes more nationalized over time, and thus the residual state-primary timing component of editing should weaken in later cycles because information production activity is increasingly organized around nationally synchronized election-season forces rather than state-specific primary calendars.*

Formally, in Model 2 (Section 4.2 Equation 3), this implies $\gamma_k < 0$ for near-primary weeks, and in particular $\bar{\gamma}_K = \frac{1}{|K|} \sum_{k \in K} \gamma_k < 0$ for the near window. Under this hypothesis, the expected pattern is that the later-year state-primary timing profile lies below the early-year profile: for each near-primary k , $f_{2024}(k) = \beta_k + 8\gamma_k < f_{2008}(k) = \beta_k$, and the gap widens as $k \rightarrow -1$. This hypothesis is disproved if $\bar{\gamma}_K$ is approximately zero (no systematic change) or positive (a strengthening state-primary timing component).

3.2. Exploratory Descriptive Analysis (EDA)

For the exploratory analysis of the editor workforce, I track each editor’s geographic and partisan scope within each election cycle. The key measure for H4 is the **state footprint** $S_{e,y}$, the number of distinct states whose House candidates editor e edited during cycle y , computed for editors with at least 5 total edits to House-candidate pages in that cycle. Formal definitions of all EDA metrics ($S_{e,y}$, $MS_{e,y}$, $\pi_{e,y}$, and related quantities) appear in Section 4.

H4 (Geographic broadening of the editor pool over time). *If the information environment surrounding House elections becomes more national in the sense emphasized by the nationalization literature (Hopkins 2018), then the organization of Wikipedia editing may also become less locally segmented. The editor workforce contributing to House-candidate pages becomes **more geographically expansive** over time.*

Concretely, I expect an upward trend (2008–2024) in: (i) the mean and median editor state footprint $S_{e,y}$, and (ii) the share of edits contributed by multi-state editors (who are characterized by $S_{e,y} \geq 3$). This hypothesis is disproved if editor footprints remain stable or become more localized, or if any observed broadening is confined to a tiny number of “super-editors” with no change in the broader editor distribution.

4. Data and Methods

4.1. Data

Wikipedia revision histories were retrieved via the MediaWiki Action API (MediaWiki 2026). On-ballot U.S. House candidates were identified by parsing Wikipedia’s election result pages for each general election from 2008 to 2024: each state’s result table contains hyperlinked candidate names, and the Wikipedia page URL embedded in each link is extracted directly, so the candidate-to-page match is definitive rather than based on name string matching. This yields a roster of candidates linked to their Wikipedia biography page titles when a page exists. Only revisions that fall within each candidate’s on-ballot two-year cycle are retained, so a single Wikipedia page may contribute to multiple election cycles if the same candidate ran multiple times. The analysis panel comprises 1,691 unique House candidate pages and 4,562 candidate-page $\times$ election-cycle observations (“page-cycles”) before censoring.

Wikipedia page presence is non-random and heavily skewed toward electorally successful candidates: 65.1% of candidate-cycles are incumbents and 79.4% are candidates who ultimately won their general election. The sample therefore systematically excludes less-notable challengers who may be most likely to exhibit sharply state-primary-timed, episodic editing activity. Results characterize the editing dynamics of *the subset of House candidates with established Wikipedia presence*, which is in essence a more nationally visible, incumbent-heavy sample, and thus should not be seen as reflective of attention on all candidates in House races in general.

The data contain 465,464 total House revisions contributed by 83,751 unique edi-

tors. Bot accounts are identified via the Wikipedia API’s user-group query (`list=users, usprop=groups`): any editor whose account belongs to the `bot` group is classified as a bot. Bot edits (3.8% of revisions) are excluded from all analyses; anonymous (IP) edits represent 22.8% of the remaining revisions. Coverage spans nine election cycles: 2008, 2010, 2012, 2014, 2016, 2018, 2020, 2022, and 2024, with page-cycles per cycle ranging from 392 (2014) to 433 (2012).

Two exclusions are applied in constructing the panel. First page-cycles where the Wikipedia page was first created within the cycle’s analysis window (after the washout) are dropped, to avoid confounding temporal patterns with page-launch dynamics; 823 page-cycles are dropped by this rule. Second, Louisiana is excluded from Q2 regression analysis because Louisiana uses a jungle primary *on the general election day* prior to 2025, making its primary dates non-comparable with other states.

The final panel includes 3,739 page-cycles and is constructed by generating the complete set of Tuesday–Monday weeks for each surviving page-cycle, assigning zero edits to weeks with no revisions, and merging state primary dates from a manually audited CSV compiled from FEC and NCSL sources. Weeks are defined as Tuesday–Monday intervals so that general election day (always a Tuesday) falls at the start of the election week, making $k = -1$ the last full week before the polls open.

The revision histories analyzed here are the complete population of edits to these pages, i.e., not a sample from a larger universe of possible edits. Confidence intervals and p-values in this paper target the *systematic component* of editing behavior: the relationship between electoral timing and editing patterns, treating individual editor decisions and race-specific events as stochastic variation around that systematic

component (one particular realization of individual actions under the same institutional environment). This is analogous to inference on administrative data, where uncertainty reflects model uncertainty rather than classical sampling variability (cf. Abadie et al. 2020).

4.2. Empirical Strategy

One goal of this study is to separate two sources of temporal variation in editing: (a) nationally synchronized election-season production dynamics, which would move all candidate pages in the same calendar weeks regardless of when their states hold primaries; and (b) state-specific electoral timing, which would produce editing concentrated around each state’s own primary calendar. U.S. general elections all occur on the same calendar date, so there is no cross-sectional variation in general-election timing to exploit. State primaries, however, are scheduled independently by each state from March through August, creating genuine cross-state variation in electoral calendar that serves as the main source of identification for the state-primary timing analysis.

The unit of analysis is a candidate-page $\times$ two-year election cycle $\times$ week (*page-cycle-week*). For each such unit, I construct two outcomes to capture editing activity:

- *Intensive margin*: $Y_{jt} = \log(1 + \text{edits}_{jt})$, measuring editing volume.
- *Extensive margin*: $P_{jt} = \mathbf{1}\{\text{edits}_{jt} > 0\}$, measuring whether any editing occurred.

The two margins capture conceptually distinct dimensions of election-season dynam-

ics. The extensive margin asks whether electoral proximity makes it more likely that *any* editing occurs on a given page in a given week, while the intensive margin asks whether, conditional on some editing occurring, the *volume* also increases. The intensive margin uses the transformed $\log(1 + \text{edits})$ because the distribution of weekly edits is highly right-skewed and zero-inflated.

Q1: election-season trajectory. To document whether editing rises in the pre-election period net of secular trends, I estimate:

$$Y_{jt} = \alpha_j + \lambda_y + \sum_{k=-34}^{-1} \beta_k^{(G)} D_k^{(G)}(j, t) + \varepsilon_{jt} \quad (1)$$

where α_j is a page-cycle fixed effect absorbing stable differences in page notability and activity level, λ_y is an election-year fixed effect absorbing secular Wikipedia-wide trends (overall editing volume declined after 2012), and $D_k^{(G)}(j, t)$ indicates that page-cycle j is exactly k weeks from the general election in week t . The regression sample consists of all page-cycle-week observations with $k \in \{-70, \dots, -1\}$, yielding 261,730 observations. Observations with $k < -70$ are treated as *washout period* to prevent updating bursts due to the prior election from contaminating the within-cycle baseline of edits, given that the focus is on pre-election editing dynamics. The $\hat{\beta}_k^{(G)}$ coefficients therefore measure the deviation of editing in week k from the baseline editing level, after absorbing stable page-cycle differences and year-level trends.

Importantly, this specification uses election-year rather than calendar-week fixed effects. Calendar-week fixed effects would absorb the nationally shared pre-election

ramp, which is the very pattern Q1 is designed to document. Election-year fixed effects absorb only between-year level shifts, leaving within-year timing variation intact in the residual.

For formal inference, I summarize the pre-election buildup using a scalar National Index,

$$NI_K = \frac{1}{|K|} \sum_{k \in K} \hat{\beta}_k^{(G)},$$

tested via a 1-df Wald test. Two windows K are reported, each motivated by a distinct feature of the U.S. congressional campaign calendar. The primary window, $K = \{-8, \dots, -1\}$, is bounded by the domestic primary schedule: by early September, *all* state primaries¹ have concluded and post-primary page-updating bursts have largely dissipated, leaving the coefficients in this window cleanly attributable to the general election period. The second, less clean window, $K = \{-10, \dots, -1\}$, corresponds to the post-Labor Day campaign sprint, the period of peak candidate advertising, field mobilization, and voter information-seeking recognized in the campaign timing literature (Fowler, Franz, and Ridout 2021; Erikson and Wlezien 2012). This second window mainly serves as a robustness check. Standard errors are two-way clustered by page-cycle and election-year.² Note that this regression documents an aggregate within-cycle pattern but cannot cleanly attribute any observed increase to electoral factors per se, as general seasonal patterns in Wikipedia activity (independent of elections) cannot be excluded without a non-candidate control group.

Q2: state-primary timing versus national synchronization. To isolate state-specific electoral timing, I exploit cross-state variation in primary dates. Let $k_{jt}^{(P)}$ denote the

signed weeks-to-primary for page-cycle j in week t . The baseline specification is:

$$\tilde{Y}_{jt}^{(P)} = \alpha_j + \delta_t + \sum_{k=-18}^{-1} \beta_k D_k^P(j, t) + \varepsilon_{jt} \quad (2)$$

where $\tilde{Y}_{jt}^{(P)} = Y_{jt} - \bar{Y}_j^{(P)}$ is editing demeaned by each page-cycle's own mean over the pre-near-primary period ($k^{(P)} \leq -19$), α_j is a page-cycle fixed effect, δ_t is a calendar-week fixed effect, and D_k^P indicates $k_{jt}^{(P)} = k$ for $k \in \{-18, \dots, -1\}$. Q2 is estimated on the intensive margin only.³

The centering $\tilde{Y}^{(P)}$ serves two purposes. First, it removes stable page-level baseline differences: because pages vary enormously in typical editing volume, a common absolute baseline would be misleading. Second, it establishes the pre-near-primary weeks ($k^{(P)} \leq -19$) as the regression's reference category: those observations are included in the estimation sample and contribute to identifying α_j and δ_t , but carry no event-time dummy. The $\hat{\beta}_k$ coefficients are therefore deviations from the pre-near-primary baseline.

The calendar-week fixed effects δ_t then absorb all nationally synchronized variation, i.e., any week-specific shock such as national news, Wikipedia-wide changes, or proximity to the general election common across all pages in that calendar week. The identifying variation is within-calendar-week: pages at different distances from their state primaries but observed in the same calendar week.⁴

To test whether this state-primary timing profile strengthens or weakens over time, a trend-interaction specification (Model 2) allows the profile to drift linearly

with election cycle:

$$\tilde{Y}_{jt}^{(P)} = \alpha_j + \delta_t + \sum_{k=-18}^{-1} (\beta_k + \gamma_k \cdot Trend_y) D_k^P(j, t) + \varepsilon_{jt} \quad (3)$$

where $Trend_y = (y - 2008)/2 \in \{0, 1, 2, 3, 4, 5, 7, 8\}$ for the eight included cycles⁵. The 2020 election cycle is excluded from Q2 because COVID-19 caused widespread, last-minute postponements of state primaries, making $k^{(P)}$ systematically mis-measured for that year; however, since the general election date was unaffected, 2020 remains in Q1. Decreasing γ_k for near-primary weeks is consistent with an attenuation of the state-primary timing component over time and thus the nationalization prediction.

For formal inference I use two separate scalar tests. **H2 (state-primary timing signal)** is tested from Model 1 via $\bar{\beta}_K^{(M1)} = \frac{1}{|K|} \sum_{k \in K} \hat{\beta}_k = 0$: this is the cleaner test because it uses the pooled estimate that averages across all eight cycles without conflating the trend parameter. **H3 (attenuation of state-primary timing)** is tested from Model 2 via $\bar{\gamma}_K = \frac{1}{|K|} \sum_{k \in K} \gamma_k = 0$. Both tests use 1-df Wald statistics with contrast vectors and two-way clustered CRV1 standard errors, and both are reported across three window widths ($K = \{-4, -3, -2, -1\}$, $K = \{-2, -1\}$, $K = \{-8, \dots, -1\}$) to assess sensitivity to window choice. Year-by-year primary-timing indices PTI_y from Model 2 are reported in Appendix E for transparency. For each page-cycle, the state primary date is matched using a manually audited CSV of state primary dates (compiled from FEC and NCSL sources). The matched primary is the state’s scheduled primary for that cycle year; for candidates who ran unopposed or lost their primary, the scheduled primary date is still used as it represents the relevant

electoral event for the state’s general-election candidates, whose pages may receive attention from the same editing community. Standard errors are two-way clustered by page-cycle and calendar-week.

The Q2 regression sample comprises $N = 247,238$ observations covering 3,316 unique page-cycles across 748 distinct calendar weeks (after excluding Louisiana and the 2020 cycle). The centering baseline ($k^{(P)} \leq -19$) contributes a median of 56 weeks per page-cycle (range: 38–78 weeks), providing a stable long-run mean for each page-cycle before the near-primary window begins. The near window contributes exactly 18 observations per page-cycle (the 18 event-time bins), and the Q2 sample is balanced: all 3,316 page-cycles contribute one observation to each of the 18 event-time bins.

EDA: characterizing the editor workforce. The regression analyses characterize *when* editing occurs relative to electoral events, but they operate at the page-cycle level and therefore do not afford insights into the composition of the workforce producing those patterns. In particular, a nationally synchronized temporal profile—if one exists—would be observationally equivalent under at least two distinct workforce structures: (i) many geographically local editors independently responding to the same national information environment, or (ii) a smaller set of geographically expansive editors whose portfolios span many states. These two structures carry different implications for how Wikipedia functions as political information infrastructure: the first suggests locally segmented production that happens to co-move, while the second suggests an integrated, cross-state editorial labor market. Distinguishing them

requires shifting the unit of analysis from the page-cycle-week to the editor-cycle and directly measuring the geographic and partisan scope of individual editing portfolios.

For each editor e and election cycle y , I construct the following measures.

Geographic scope. The **state footprint**

$$S_{e,y} = |\{s : e \text{ edited at least one House-candidate page in state } s \text{ during cycle } y\}|$$

counts the number of distinct states whose candidates editor e edited. The **modal-state share** $MS_{e,y} = \max_s n_{e,s,y} / N_{e,y}$ is the fraction of editor e 's cycle- y edits directed to this editor's most-edited state, where $n_{e,s,y}$ is the edit count to state s and $N_{e,y} = \sum_s n_{e,s,y}$ is the total; higher values indicate greater geographic concentration. Editors with $S_{e,y} \geq 3$ are classified as *multi-state editors*. All geographic measures are restricted to editors satisfying the activity threshold $N_{e,y} \geq 5$, to exclude one-off contributors.

Partisan scope. Let $N_{e,y}^D$ and $N_{e,y}^R$ denote editor e 's edit counts to Democratic and Republican candidates in cycle y , respectively. An editor is a *cross-party editor* if $N_{e,y}^D > 0$ and $N_{e,y}^R > 0$. For continuous measurement, **party concentration** is summarized by the Laplace-smoothed index $\pi_{e,y} = (N_{e,y}^D + 1) / (N_{e,y}^D + N_{e,y}^R + 2)$, which maps to $(0, 1)$ with values near 0 indicating Republican-specialist editors, near 1 indicating Democratic-specialist editors, and near 0.5 indicating balanced cross-party editing; the smoothing prevents mechanical boundary spikes for editors with very few partisan edits. Partisan measures are computed solely for editors with $N_{e,y}^D + N_{e,y}^R \geq 5$.

Composition-shift analysis. To assess whether the pre-election ramp changes the

type of editing being produced, the same geographic and partisan scope measures are computed separately within two time windows: the **baseline** (the portion of the analysis window outside the 18-week pre-general-election period, excluding the washout) and the **near-general** window ($k^{(G)} \in \{-18, \dots, -1\}$). Comparing cross-party shares and geographic breadth between these windows reveals whether the editing that intensifies before the general election differs in composition from the editing produced during the rest of the cycle.

All measures describe the geographic and partisan scope of *editing portfolios*, not editors’ physical locations or political affiliations. Registered users’ physical locations are hard to track due to not only the potential use of IP masking, but also the fact that their IP addresses are only accessible by CheckUsers, a small group of trusted Wikipedia users. The EDA is explicitly exploratory and reports descriptive patterns without formal hypothesis tests, with a goal of providing mechanistic interpretation for the regression findings.

5. Regression Results

5.1. Q1. Election-Cycle Time Patterns in Candidate-Page Editing

Figure 1 summarizes within-page-cycle editing dynamics over the two-year general-election cycle using raw mean outcomes. Panel A displays the full two-year trajectory. The dominant feature is a large burst of post-election page updating immediately following each general election; this is an order of magnitude larger than any pre-election variation. This asymmetry suggests that the single most important driver of

editing on candidate pages is the arrival of verifiable new facts (election outcomes, office-holding status, committee assignments) rather than anticipatory information production in the run-up to the vote. The shaded region in Panel A ($k = -52$ to $k = -1$) is expanded in Panels B and C to examine the pre-election period more closely.

Panels B and C display the intensive margin $[\log(1 + \text{edits})]$ and the extensive margin [probability of any editing], respectively, aggregated to 4-week windows to smooth week-to-week noise. The horizontal dashed line marks the reference-period average ($k \in \{-70, \dots, -35\}$), a mid-cycle window well before primary season begins. Against this baseline, both margins show a gradual upward trajectory as the general election approaches, with activity rising visibly above the reference level in the post-primary window ($k \in \{-8, \dots, -1\}$). Dotted vertical lines bound the approximate primary season (March through August). Notably, some increase is already apparent during the primary season itself, raising the question of whether this mid-year activity tracks state-specific primary calendars or reflects a broader national trend.

To formally test whether the pre-election increase is distinguishable from zero after absorbing stable page-cycle differences and secular year-to-year trends, I estimate Equation (1). Figure 2 plots the resulting coefficient profile $\hat{\beta}_k^{(G)}$ for $k \in \{-34, \dots, -1\}$. The profile reveals an instructive pattern. Before the post-primary window ($k \leq -9$), coefficients fluctuate around zero but with notable positive excursions at approximately 19 and 14 weeks before the general election, roughly coinciding with the heart of primary season. These mid-year spikes could reflect bursts of editing activity triggered by specific primaries, contested nominations, or surges in media

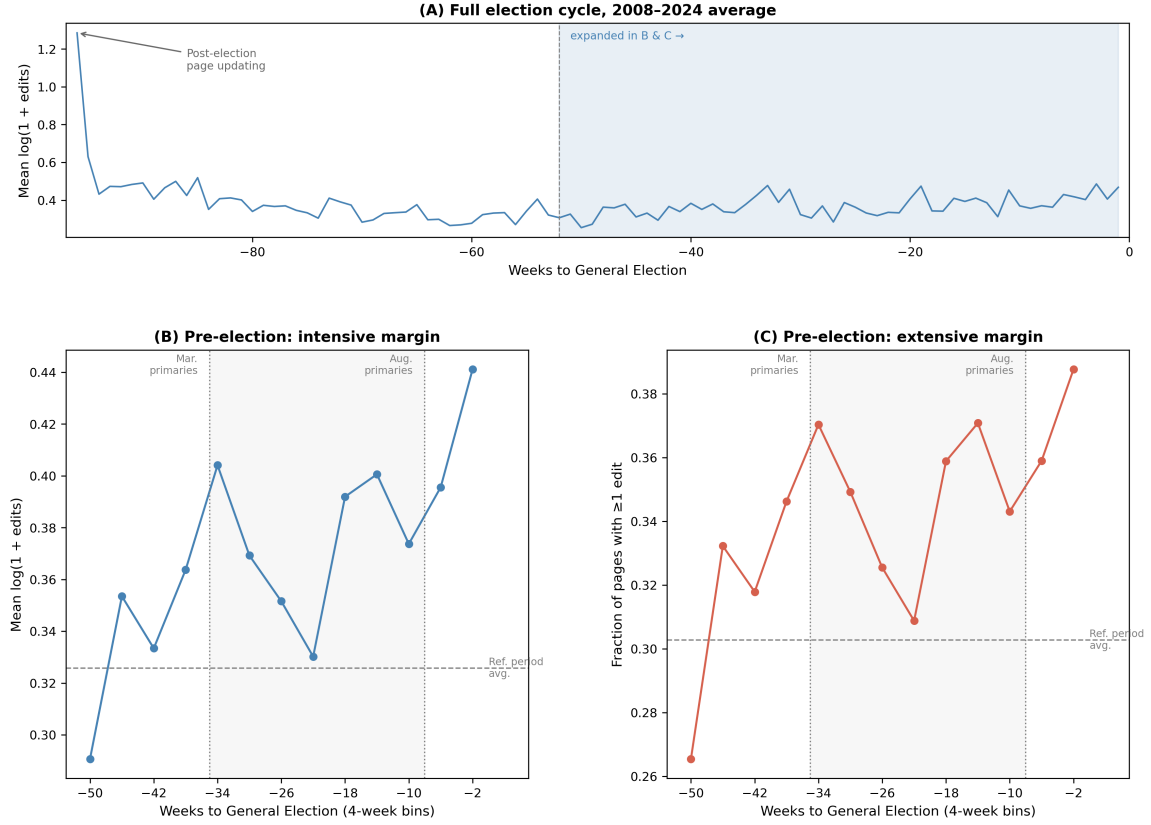


Figure 1. Wikipedia Editing on U.S. House Candidate Pages over the Election Cycle (2008–2024 Pooled Average).

Note: Panels B and C display the intensive margin $[\log(1 + \text{edits})]$ and the extensive margin [probability of any editing], respectively, during the pre-election period.

Each point is the mean across a 4-week window. The horizontal dashed line marks the reference-period average ($k \in \{-70, \dots, -35\}$, the mid-cycle baseline level edits). Dotted vertical lines bound the approximate primary season (March - August). Any pre-election increase in the final weeks is tested formally in Table 1.

coverage surrounding particular races, though the Q1 specification cannot distinguish among these possibilities. Within the post-primary window ($k \in \{-8, \dots, -1\}$), the profile shows a more sustained upward trajectory across both margins, rising steadily as the general election approaches. Because coefficients for weeks during primary season may conflate pre-general-election signal with post-primary information updating, the formal hypothesis test is restricted to two windows: the primary window $k \in \{-8, \dots, -1\}$, chosen because all state primaries have concluded by this point; and a robustness window $k \in \{-10, \dots, -1\}$, corresponding to the post-Labor Day campaign sprint recognized in the campaign timing literature (Fowler, Franz, and Ridout 2021; Erikson and Wlezien 2012). Table 1 reports the corresponding scalar tests.

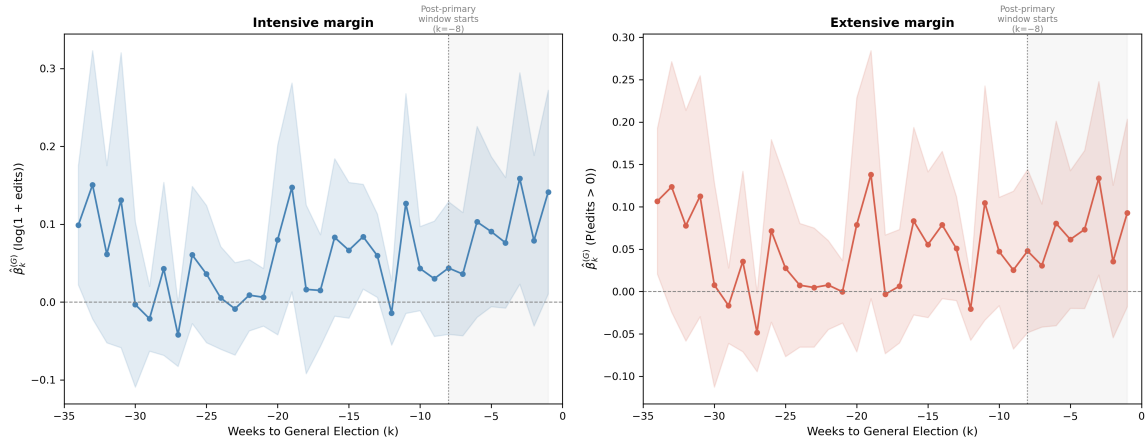


Figure 2. Q1 Event-Study Coefficient Profile.

Note: $\hat{\beta}_k^{(G)}$ for $k \in \{-34, \dots, -1\}$, with 95% confidence intervals.

Left panel: intensive margin $[\log(1 + \text{edits})]$. Right panel: extensive margin $[\Pr(\text{edits} > 0)]$, as compared to the reference level.

The horizontal dashed line at zero represents the average editing level during the reference window $k \in \{-70, \dots, -35\}$.

FE: page-cycle and election-year. SE: two-way clustered.

Across both outcome measures and both windows, NI estimates are positive

Table 1. Q1 National Index (NI_K) Scalar Tests.

Outcome	Window	NI	SE	Wald	p-value
Intensive (log edits)	$k \in \{-8, \dots, -1\}$	0.0911	0.0347	6.89	0.009***
Intensive (log edits)	$k \in \{-10, \dots, -1\}$	0.0802	0.0305	6.90	0.009***
Extensive ($P > 0$)	$k \in \{-8, \dots, -1\}$	0.0695	0.0286	5.91	0.015**
Extensive ($P > 0$)	$k \in \{-10, \dots, -1\}$	0.0629	0.0263	5.72	0.017**

Note: $NI_K = \frac{1}{|K|} \sum_{k \in K} \hat{\beta}_k^{(G)}$.

FE: page-cycle + election-year. SE: two-way clustered. $N = 261,730$.

*** $p < 0.01$; ** $p < 0.05$. (1-df Wald test, with a single linear restriction $H_0 : NI_K = 0$)

and statistically distinguishable from zero (Table 1). For the post-primary window $K = \{-8, \dots, -1\}$, editing volume rises by roughly 0.09 log-units above the mid-cycle baseline; the extensive-margin estimate is $NI = 0.070$ ($SE = 0.029$, $p = 0.015$), indicating that the probability of any editing in a given week rises by approximately 7 percentage points. The post-Labor Day window $K = \{-10, \dots, -1\}$ yields consistent magnitudes ($NI = 0.080$ and 0.063 for the intensive and extensive margins, respectively; both $p < 0.05$). More concretely, the $NI = 0.091$ intensive-margin estimate corresponds to moving from approximately 0.39 expected edits per page-week at the mid-cycle baseline to 0.52 in the pre-election period, a roughly 34% increase in the absolute editing rate.

This pre-election ramp is real but modest, especially when set against the post-election updating burst visible in Panel A. The contrast suggests that Wikipedia’s primary function around elections is retrospective documentation rather than prospective information preparation for voters. Nevertheless, the pre-election increase is consistent with some combination of the reactive supply and electoral engagement mechanisms introduced in Section 1: as the general election approaches, the volume

of documentable political news increases and editors’ attentiveness to candidate pages rises, both producing more revisions. An important caveat, however, is that this specification alone cannot attribute the observed increase to electoral factors per se. For instance, fall-semester university courses routinely assign Wikipedia biography editing as coursework, producing a seasonal influx of edits to biographical articles that overlaps with the pre-election window but is unrelated to electoral dynamics; without a non-candidate control group, such co-occurring seasonal patterns cannot be ruled out.

5.2. Q2. *State-Primary Timing versus National Synchronization*

Figure 3 plots $\hat{\beta}_k$ for $k \in \{-18, \dots, -1\}$ from Model 1. Each coefficient corresponds to a within-calendar-week comparison, i.e., among all page-cycles observed in the same calendar week, how much more editing a page receives when it is k weeks from its state primary, relative to pages that are more than 19 weeks from theirs. Calendar-week fixed effects absorb all nationally synchronized variation, so $\hat{\beta}_k$ isolates the residual state-primary timing component, the part of editing activity that follows the state’s own electoral calendar over and above the national rhythm. If state primaries organize editorial production, the profile should show a systematic upward gradient peaking in the weeks just before the primary.

Figure 3 does not show this pattern. Estimates fluctuate narrowly around zero across all 18 event-time bins with no consistent upward gradient. Thus, while candidate-page editing does intensify before elections, that intensification appears to move in common across states. The finding is consistent with what the nationalization litera-

ture would predict for the information environment surrounding House races (Hopkins 2018; Moskowitz 2021); even production-side activity on a platform as decentralized as Wikipedia is organized around a national electoral rhythm.

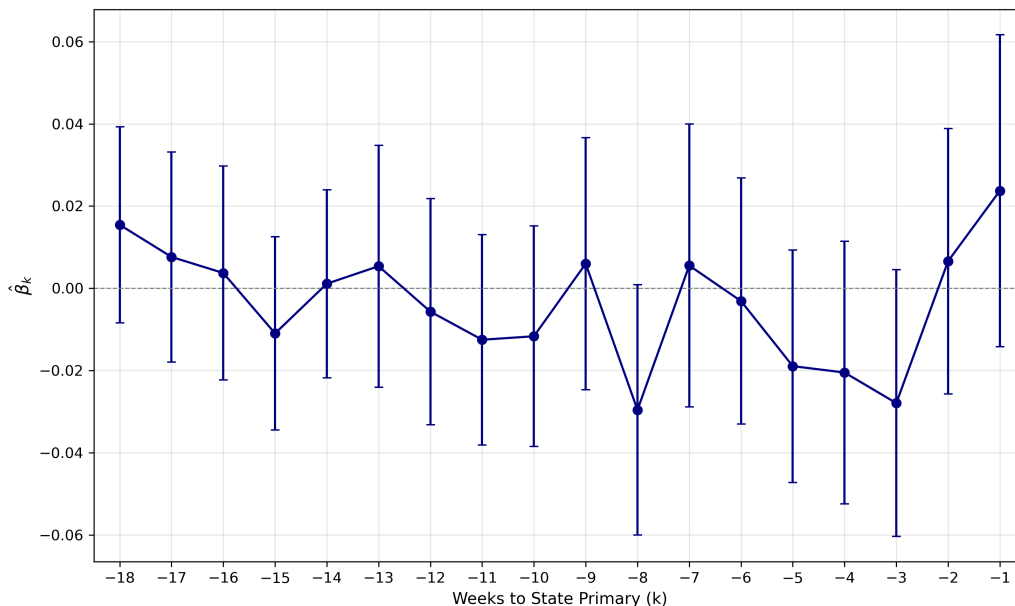


Figure 3. Q2 State-Primary Timing Profile from Model 1.

Note: $\hat{\beta}_k$ for $k \in \{-18, \dots, -1\}$ with 95% confidence intervals. The precise interpretation of each coefficient is: within a given calendar week, by how much editing on a page k weeks before its state primary differs from a page that is more than 19 weeks before its primary, after removing each page’s own long-run pre-primary mean.

The baseline (zero) is the pre-near-primary period ($k^{(P)} \leq -19$), which serves as both the centering reference for the outcome and the omitted regression category. The horizontal dashed line at zero is the null of no state-primary timing effect, and represents the absence of any state-specific primary-timing signal net of national synchronization.

SE: two-way clustered by page-cycle and calendar-week.

5.3. Q2 over Time: Change in the State-Primary Timing Profile (2008 vs. 2024)

A natural follow-up is whether the absence of a state-primary timing signal is a recent development, perhaps reflecting the progressive nationalization of the information environment, or whether it characterizes the entire 2008–2024 period. Model 2 addresses this by allowing the state-primary timing profile to drift linearly with

the election cycle ($Trend = 0$ in 2008, $Trend = 8$ in 2024). Figure 4 visualizes the implied profiles for 2008 and 2024. To illustrate the scale of the estimated change: for the week immediately before the primary ($k = -1$), the estimates imply $\beta_{-1} = 0.0323$ and $\gamma_{-1} = -0.0023$, yielding an implied 2024 effect of $f(-1, 2024) = \beta_{-1} + 8\gamma_{-1} = 0.0138$ —a change that is small relative to the confidence interval width. Neither endpoint shows a clear state-primary timing signal, and the two profiles do not display a systematic downward shift consistent with progressive attenuation.

Scalar Wald tests avoid the multiple-testing problem inherent in examining 18 individual coefficients. Table 2 reports the H2 test: the average near-primary effect $\bar{\beta}_K^{(M1)}$ from Model 1 is close to zero under all three window widths, yielding no statistically distinguishable state-primary timing signal. Table 3 reports the H3 test: the average trend coefficient $\bar{\gamma}_K$ from Model 2 is likewise indistinguishable from zero, providing no evidence that the state-primary timing component has weakened (or strengthened) over time. Year-by-year primary-timing indices (PTI_y) from Model 2 are reported in Appendix Table 5 for transparency; all eight cycle-specific estimates are individually indistinguishable from zero.

Together, the H2 and H3 results suggest that the absence of state-primary timing is a persistent feature of candidate-page editing throughout the study period. Even in 2008, when the nationalization of House elections was arguably less advanced than in later cycles, editing activity was already organized around a common national calendar. The implication is that Wikipedia’s production process for candidate biographies may have been nationally structured from the outset of the period studied,

rather than having gradually lost a local character over time.

A legitimate concern here is whether the null reflects a true absence of state-primary timing effects or simply insufficient statistical power. Given the standard error of 0.0132 for $\bar{\beta}_K^{(M1)}$ in the main window (Table 2), the minimum detectable effect at 80% power and $\alpha = 0.05$ is approximately 0.037 log-units.⁶ This threshold is well below the $NI = 0.091$ from Q1, which suggests that a state-primary timing effect of comparable magnitude to the general-election buildup would have been detectable. The near-zero estimates are therefore informative nulls.

Table 2. Test for H2: Average Near-Primary State-Primary Timing Effect from Model 1.

Window K	$\bar{\beta}_K^{(M1)}$	SE	Wald	p-value
$\{-4, -3, -2, -1\}$ (main)	-0.0045	0.0132	0.1180	0.7312
$\{-2, -1\}$ (narrower)	0.0152	0.0154	0.9755	0.3233
$\{-8, \dots, -1\}$ (wider)	-0.0080	0.0112	0.5118	0.4744

Note: $\bar{\beta}_K^{(M1)} = \frac{1}{|K|} \sum_{k \in K} \hat{\beta}_k$ from Model 1 (Equation 2). The 2020 cycle is excluded due to COVID-19 disruptions to state primary scheduling. $N = 247,238$ (3,316 page-cycles, 748 calendar weeks).

$H_0: \bar{\beta}_K^{(M1)} = 0$, tested via 1-df Wald test. SE: two-way clustered (page-cycle $\times$ calendar-week).

Table 3. Test for H3: Attenuation of State-Primary Timing from Model 2.

Window K	$\bar{\gamma}_K$	SE	Wald	p-value
$\{-4, -3, -2, -1\}$ (main)	-0.0018	0.0053	0.1121	0.7377
$\{-2, -1\}$ (narrower)	-0.0018	0.0062	0.0830	0.7733
$\{-8, \dots, -1\}$ (wider)	-0.0023	0.0044	0.2635	0.6077

Note: $\bar{\gamma}_K = \frac{1}{|K|} \sum_{k \in K} \hat{\gamma}_k$ from Model 2 (Equation 3). Same sample and data treatment as in Table 2.

$H_0: \bar{\gamma}_K = 0$, tested via 1-df Wald test; SE: two-way clustered (page-cycle $\times$ calendar-week).

Several specification checks confirm the robustness of these results. Both the H2 and H3 tests are reported under narrower ($K = \{-2, -1\}$) and wider ($K =$

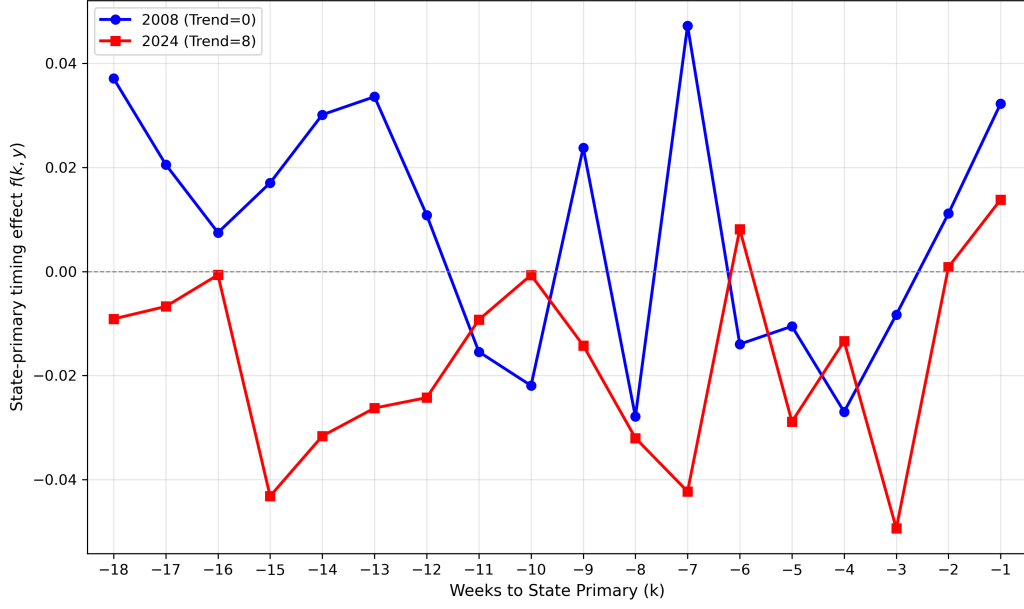


Figure 4. Change in the State-Primary Timing Profile over Time (Model 2).

Note: Lines plot the implied state-primary timing profile $f(k, y) = \beta_k + \gamma_k \cdot Trend_y$ from Model 2 for 2008 ($Trend = 0$) and 2024 ($Trend = 8$). The zero on the y -axis represents the null of no state-primary timing effect (no state-specific primary-timing deviation net of national synchronization and page-cycle baseline). A value above zero at some k means pages at that distance from their primary show systematically more editing than comparable pages in the same calendar week that are far from their primary.

$\{-8, \dots, -1\}$) near windows alongside the main specification in Tables 2 and 3; results are qualitatively stable across all three window widths, with no window yielding a statistically distinguishable effect. States with nonstandard primary institutions (top-two systems) are retained in the main specification because their primary dates are still identified; excluding them does not meaningfully alter the results. Splitting the sample by editor type (registered vs. anonymous) is examined in the descriptive analysis (Section 6) only, given the small IP sample size.

6. Descriptive Results

The regression analyses characterize *when* editing occurs relative to electoral events, but they operate at the page-cycle level and are therefore silent on the composition of the workforce producing those patterns. In particular, a nationally synchronized temporal profile is observationally equivalent under at least two distinct workforce structures: (i) many geographically local editors independently responding to the same national information environment, or (ii) a smaller set of geographically expansive editors whose portfolios span many states. These two structures carry different implications for how Wikipedia functions as political information infrastructure: the first suggests locally segmented production that happens to co-move, while the second suggests an integrated, cross-state editorial labor market. Distinguishing them requires shifting the unit of analysis from the page-cycle-week to the editor-cycle and directly measuring the geographic and partisan scope of individual editing portfolios. All measures below describe the scope of editing portfolios, not editors' physical locations or political affiliations.

6.1. Geographic Breadth of Editor Portfolios

The following results focus on registered accounts. IP editors⁷ passing the $N_{e,y} \geq 5$ activity threshold are far fewer per cycle (142 to 591, compared to 624 to 1,576 for registered editors), yielding substantially higher sampling variance; their results are shown alongside registered editors in the figures in Appendix C for completeness.

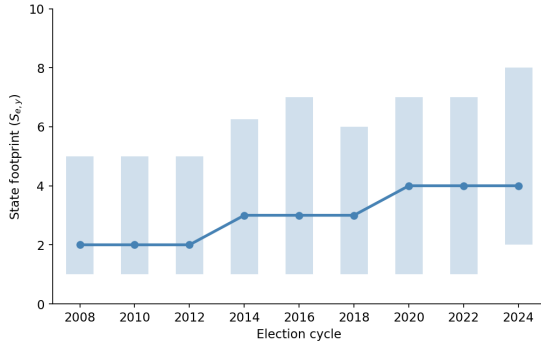


Figure 5. State Footprint by Election Cycle.

Note: $S_{e,y}$ = number of distinct states in which editor e edited at least one House-candidate page during cycle y . Registered editors with $N_{e,y} \geq 5$. Shaded bars show the IQR; connected line shows the cycle median. Sample sizes per cycle: 2008 $n=1,576$; 2010 $n=1,278$; 2012 $n=1,049$; 2014 $n=752$; 2016 $n=624$; 2018 $n=1,054$; 2020 $n=1,224$; 2022 $n=1,172$; 2024 $n=1,112$.

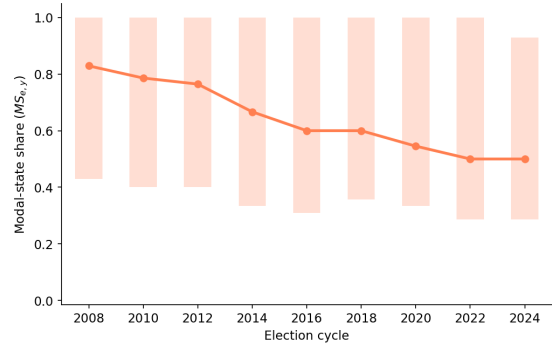


Figure 6. Modal-State Share by Election Cycle.

Note: $MS_{e,y}$ = fraction of editor e 's cycle- y edits directed to this editor's most-edited state. Registered editors with $N_{e,y} \geq 5$. Shaded bars show the IQR; connected line shows the cycle median. Higher values indicate greater geographic concentration within one state. Sample sizes are identical to Figure 5.

Figure 5 reveals two concurrent features. First, a persistent mass of single-state editors remains present across all cycles, indicating that geographically local editing never disappears. Second, the center of the distribution shifts upward over time: the median state footprint rises from 2.0 (2008–2012) to 4.0 (2020–2024), and the mean increases from 4.5 to 6.5. The right tail remains long throughout, with a small number of editors contributing across dozens of states. Importantly, the broadening occurs

in the middle of the distribution, not solely in the tails, suggesting that growth in multi-state activity reflects a general shift in how editors engage with candidate pages rather than the entry of a few prolific outliers. It is also worth noting that the number of qualifying editors ($N_{e,y} \geq 5$) is non-monotonic across cycles, declining from 1,576 in 2008 to a trough of 624 in 2016 before rebounding to over 1,000 in the 2018–2024 cycles. The geographic broadening occurs despite this contraction and recovery in the editor pool; thus, this likely reflects a genuine change in editing behavior.

Figure 6 provides a complementary perspective by measuring within-state concentration. The distribution of $MS_{e,y}$ shifts downward over time, confirming that the expanding geographic footprint is not merely an artifact of editors adding occasional single edits in additional states: the share of effort directed to any one state declines as editors spread their contributions more evenly across states.

Figure 7 distills these distributional shifts into a single time series. Both the editor-weighted and edit-weighted multi-state shares ($S_{e,y} \geq 3$) increase over time, but the edit-weighted share is substantially higher throughout, roughly 70% in 2008 rising to nearly 90% by 2024. The persistent gap between the two series indicates that multi-state editors are not just more numerous over time but are systematically higher-volume contributors. In other words, the editors who produce the most content on candidate pages are disproportionately those whose portfolios span many states. This concentration of editing volume among geographically broad contributors helps explain the Q2 null result: if the editors who dominate the revision count are attending to candidates across many states simultaneously, their activity would co-move with the national electoral calendar.

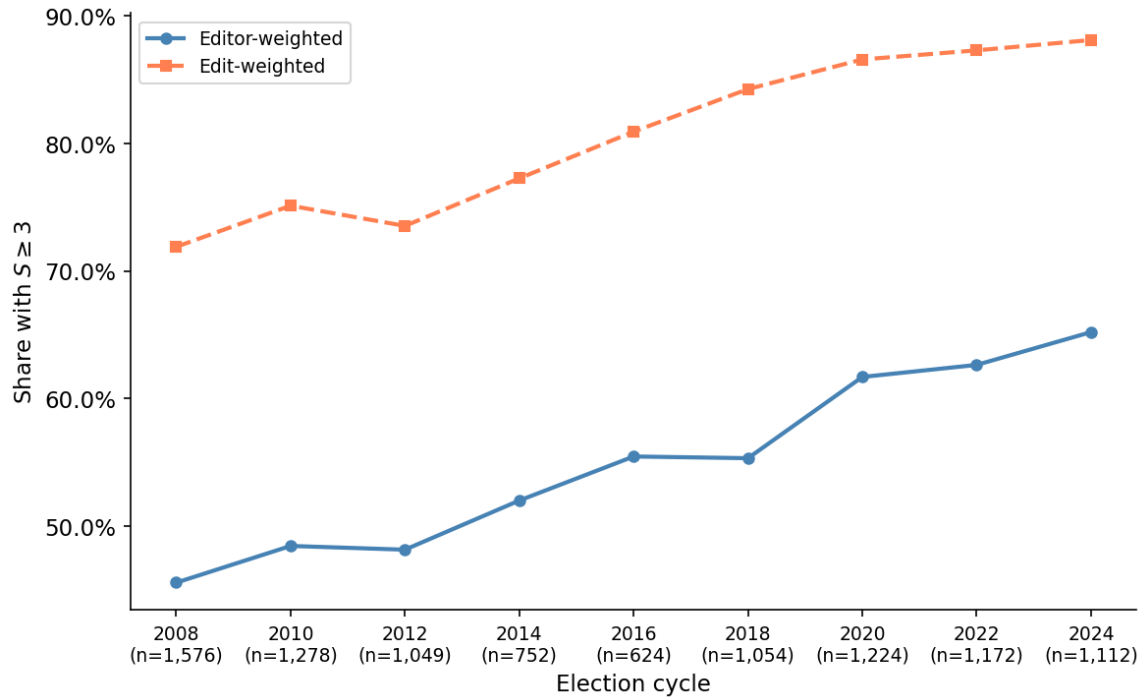


Figure 7. Multi-State Editor Share over Election Cycles.

Note: Each point shows the fraction of editors (editor-weighted) or fraction of total edits (edit-weighted) attributable to editors with $S_{e,y} \geq 3$ (multi-state editors). Registered editors with $N_{e,y} \geq 5$. The persistent gap between the edit-weighted and editor-weighted series indicates that multi-state editors contribute disproportionately large editing volume.

6.2. Cross-Party Participation by Cycle

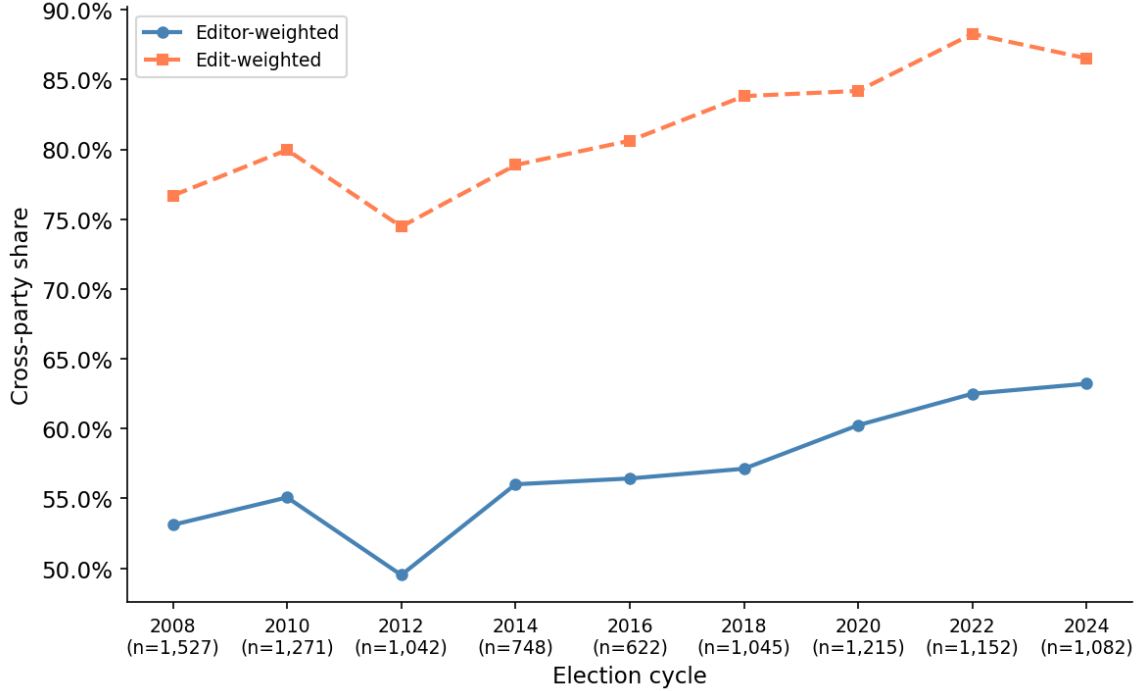


Figure 8. Cross-Party Participation by Election Cycle.

Note: Each point shows the fraction of editors (editor-weighted) or fraction of total partisan edits (edit-weighted) from editors who edited both Democratic ($N_{e,y}^D > 0$) and Republican ($N_{e,y}^R > 0$) candidates in the cycle. Restricted to editors with $N_{e,y}^D + N_{e,y}^R \geq 5$. The persistent gap between edit-weighted and editor-weighted series indicates that higher-volume contributors are more likely to edit across party lines.

Figure 8 reports cross-party participation among editors with at least 5 partisan edits ($N^D + N^R \geq 5$). Two patterns stand out. First, cross-party participation is systematically higher on an edit-weighted basis than on an editor-weighted basis, echoing the geographic finding: the highest-volume contributors are disproportionately those who edit candidates of both parties. Second, both series trend upward over the study period, with the edit-weighted cross-party share rising from roughly 70% to over 80%. This is consistent with the geographic broadening documented above, and suggests a progressively more integrated editing workforce in which the dominant contributors

operate as nonpartisan generalists rather than single-party specialists.

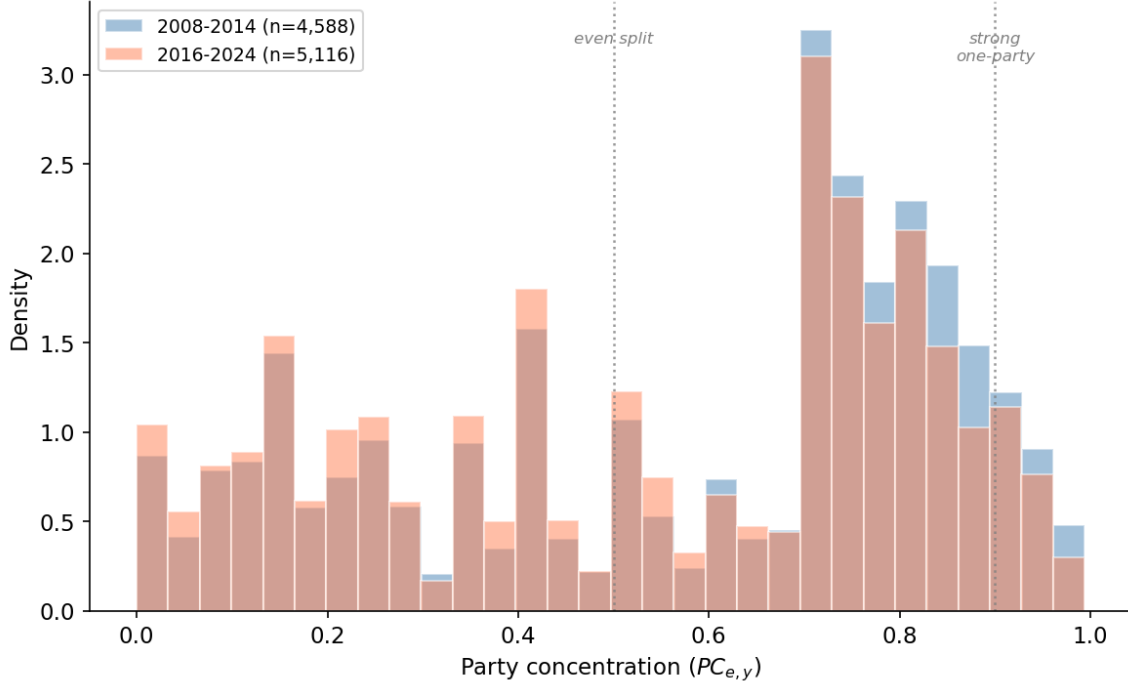


Figure 9. Distribution of Party Concentration Index.

Note: $\pi_{e,y} = (N_{e,y}^D + 1) / (N_{e,y}^D + N_{e,y}^R + 2)$ across editors, pooled into early (2008–2016) and late (2018–2024) periods. Values near 0 indicate Republican-specialist editors; near 1 indicate Democratic-specialist editors; near 0.5 indicate balanced cross-party editing. Restricted to editors with $N_{e,y}^D + N_{e,y}^R \geq 5$. Additive smoothing (Laplace) prevents mechanical spikes at the boundaries for very low-edit contributors.

Figure 9 further provides granularity in information by examining the full distribution of party concentration beyond the binary cross-party threshold. Pooling cycles into early (2008–2016) and late (2018–2024) periods, the distributions appear remarkably stable. A substantial mass remains at the partisan extremes ($\pi_{e,y}$ near 0 or 1), and the central peak near 0.5 does not shift appreciably between periods. At first glance, this stability may seem to contradict the rising cross-party share in Figure 8. The reconciliation is that the cross-party threshold ($N^D > 0$ and $N^R > 0$) is a coarse binary indicator, sensitive to whether an editor makes even a single edit to the

other party; the continuous party concentration index captures the *degree* of partisan balance, which has not changed much. The implication is that while more editors are crossing party lines at the margin, the overall structure of partisan specialization—with some editors working predominantly on one party’s candidates and others editing both—has remained stable over the past decade and a half. Wikipedia’s political editing workforce is not converging toward a uniformly balanced composition; rather, a persistent mix of specialists and generalists coexists, with the generalists commanding a growing share of total volume.

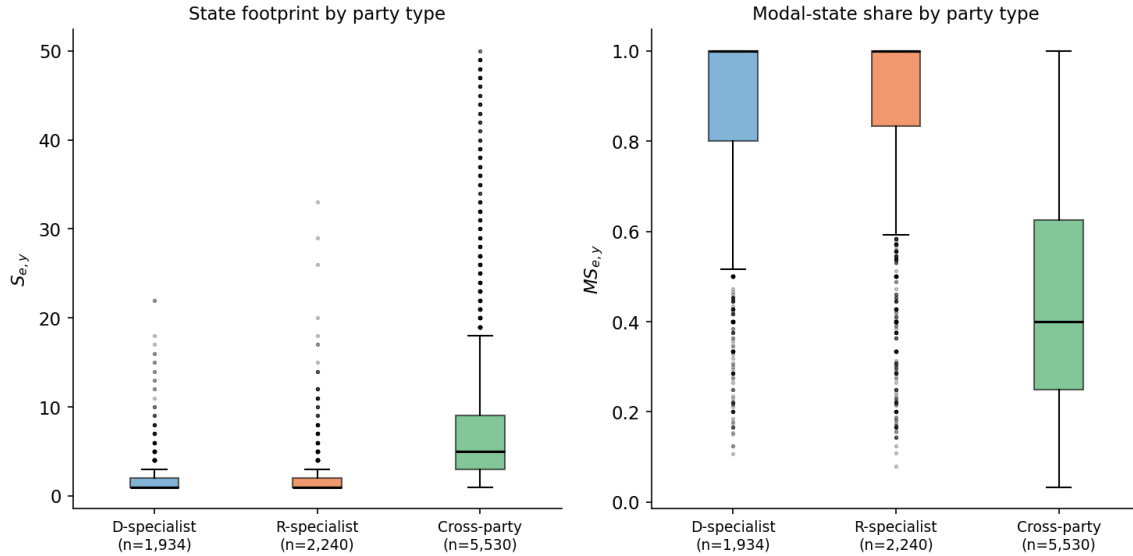


Figure 10. Geographic Scope by Party Type (Pooled across Cycles).

Note: State footprint $S_{e,y}$ (left) and modal-state share $MS_{e,y}$ (right) by party type. Party type: cross-party ($N_{e,y}^D > 0$ and $N_{e,y}^R > 0$), Democratic-specialist ($N_{e,y}^D > 0$, $N_{e,y}^R = 0$), or Republican-specialist ($N_{e,y}^D = 0$, $N_{e,y}^R > 0$). Restricted to editors with $N_{e,y}^D + N_{e,y}^R \geq 5$.

Figure 10 provides perhaps the most revealing characterization of the editing workforce by linking partisan scope directly to geographic scope. Cross-party editors have larger geographic reach (mean $S_{e,y} \approx 8.15$, median = 5) and lower within-state concentration (mean $MS_{e,y} \approx 0.46$). In contrast, Democratic- and Republican-

specialist editors are overwhelmingly single-state (median $S = 1$) and highly concentrated within that state (modal-state shares near 1). This tight association between cross-party editing and cross-state editing is difficult to reconcile with a model of partisan actors strategically managing candidate pages. An editor who systematically edits candidates of both parties across many states looks far more like a maintenance-oriented generalist than like an advocate for one party’s candidates. The revisions for such maintenance include updating infoboxes and standardizing formatting in addition to adding sourced information. The pattern aligns with the “power of the few” finding in the peer production literature, where a small core of prolific contributors produces a disproportionate share of content (Kittur et al. 2007), and with the role differentiation documented in studies of Wikipedia’s editing ecology (Müller-Birn et al. 2015). What this analysis adds is that on political pages specifically, the prolific core is not only high-volume but also expansive in geographic and partisan scope; in other words, these contributors appear to be informational caretakers of the candidate-page infrastructure.

6.3. Composition Shift as General Election Approaches

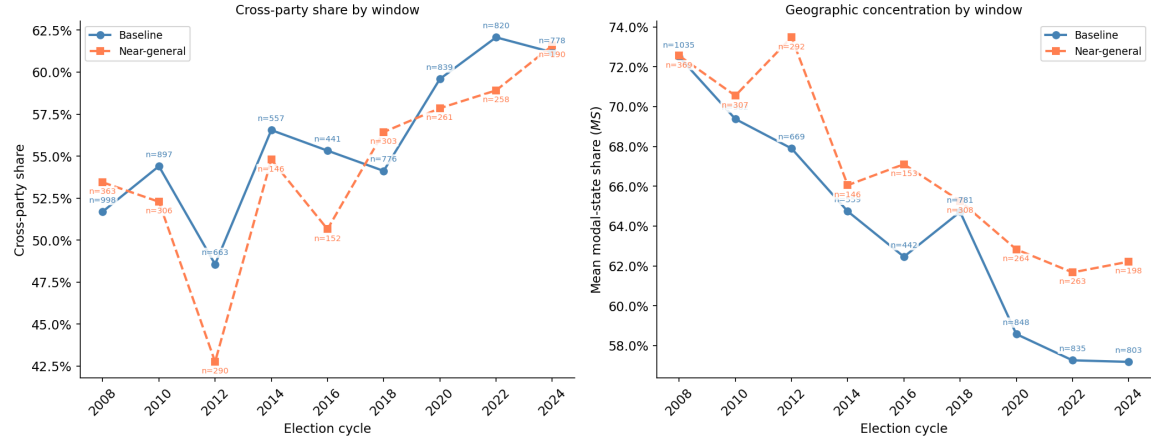


Figure 11. Cross-Party Share and Geographic Concentration by Time Window. *Note:* Among registered editors with $N_{e,y}^D + N_{e,y}^R \geq 5$ (left) or $N_{e,y} \geq 5$ (right) in each window. Baseline: portion of the analysis window outside the 18-week pre-general-election period (excluding post-election updating). Near-general: 18 weeks before the general election ($k^{(G)} \in \{-18, \dots, -1\}$). Left panel: fraction of editors whose edits span both parties. Right panel: mean modal-state share $MS_{e,y}$ (fraction of an editor’s edits going to their most-edited state), a proportion that is invariant to window length. n = number of unique editors passing the respective threshold in that cycle $\times$ window.

The regression results in Q1 establish that editing intensifies before the general election. One may wonder whether this pre-election ramp reflects the entry of a different *type* of editor, perhaps more partisan, more geographically local, or otherwise distinct from the year-round editing workforce. If so, the pre-election period could represent a qualitatively different production regime even if the aggregate volume change is modest. I therefore compare the cross-party share of editing between the baseline and near-general-election windows ($k^{(G)} \in \{-18, \dots, -1\}$) in Figure 11.

On the partisan dimension, the composition is relatively stable; across all nine cycles, the baseline cross-party share averages 0.559 and the near-general share averages 0.543, a difference of only -0.016 that is inconsistent in sign across cycles. This

stability is evidence against the entry of a distinctively partisan editing population during the pre-election window.

The results on geographic dimension suggest more complicated patterns. I use the modal-state share $MS_{e,y}$ —the fraction of an editor’s edits going to their most-edited state—rather than the raw state count $S_{e,y}$, because MS is a proportion that does not mechanically increase with window length.⁸ Mean MS shifts modestly upward in the near-general window (baseline $\overline{MS} = 0.639$, near-general $\overline{MS} = 0.669$ with a shift of +0.030), and this increase is positive in all nine cycles. Near-general editors are slightly *more* concentrated in their top state than the year-round editing workforce. This pattern could be explained by two non-exclusive underlying forces: the same cross-state generalists may tilt their effort toward home-state candidates as general elections for their state representatives become more salient, or some modestly more geographically focused editors may enter the editing process during the pre-election window.

The two panels suggest that the pre-election ramp is primarily a change in *intensity* rather than in *kind*, but with a geographic qualification. Partisan composition does not shift, suggesting the absence of a systematic entry of strategic single-party actors. Geographic concentration increases slightly, but with a modest magnitude (3 percentage points on a 0–1 scale), and at this increased concentration, editors in the near-general window are still distributing roughly a third of their edits across non-modal states ($\overline{MS} = 0.669$). The pre-election ramp does not bring a fundamentally different editing workforce to candidate pages, but it may modestly intensify editors’ focus on their primary-state candidates.

7. Discussion

The central finding is that the production of political information on Wikipedia, at least for U.S. House candidate biography pages, is organized around a national electoral rhythm rather than around the state-specific institutional calendars that formally structure American primary elections. In what follows, I further expand on the implications of specific results provided by the different analytical approaches.

7.1. *Wikipedia as a Retrospective Archive*

The most visually striking feature of the election-cycle trajectory (Figure 1, Panel A) is the dominance of post-election updating over pre-election production. This asymmetry reveals that, within the political information ecosystem, the platform’s primary function around elections is still *retrospective documentation*, i.e., recording verified outcomes, updating office-holding status, and incorporating post-election reporting into candidate biographies. This aligns with Wikipedia’s institutional commitment to verifiability and reliable sourcing; such policies inherently favor documenting established facts over anticipating future developments.

The pre-election ramp documented in Q1, while statistically distinguishable from zero, is modest by comparison. Whether it reflects a genuine election-season intensification or co-occurring seasonal patterns in Wikipedia activity (such as fall-semester coursework in universities that assigns biography editing) cannot be determined without a non-candidate control group. Nevertheless, its existence and its national rather than state-specific character are consistent with a combination of the *reactive supply*

and *electoral engagement* mechanisms introduced in Section 1. As the general election approaches, the volume of documentable political news increases and editors' attentiveness to candidate pages rises, thus producing more revisions. This pre-election activity, however motivated, does not change the partisan composition of editing; the same mix of cross-party generalists and single-party specialists operates before and during the pre-election window. Geographic concentration increases slightly, with editors focusing modestly more on their top state, but the shift is small and does not alter the dominant cross-state character of the editing workforce.

7.2. *National Synchronization Thesis*

The Q2 null result is important for the American politics literature on nationalization. U.S. House primaries are formally organized at the state level, with primary dates spread across a five-month window from March through August. If the editing community closely tracked these local institutional calendars, which respond to state-specific nomination contests and local media coverage of primary campaigns, the state-primary timing profile would show a systematic upward gradient as each page approached its own state's primary. The flat profile in Figure 3, combined with the near-zero scalar tests across all window choices, indicates otherwise.

This finding has an analog in the nationalization literature's account of how political attention is structured but should be interpreted with caution. Put precisely, Hopkins (2018) shows that citizens increasingly encounter politics through nationally salient partisan frames rather than through state- and locality-specific contexts, whereas Moskowitz (2021) demonstrates that local television news can interrupt this

process by supplying information that enables more independent evaluations of gubernatorial and senatorial candidates. However, instead of bearing directly on traditional media exposure or political cognition, my findings at most reveal a pattern on the temporal organization of political attention. The weak correspondence between Wikipedia editing rhythms and state primary calendars is suggestive and is broadly *compatible* with the claim that attention to down-ballot actors is often organized by forces other than locally bounded electoral events.

A key limitation of this analysis, however, is that it only captures when candidate pages are edited, but not what those edits are substantively about. The temporal decoupling from state electoral events could indicate a broader nationalization of political attention, but it could also arise from platform-internal dynamics that have little to do with how candidates are politically interpreted. This could also help explain the absence of a trend in *progressive attenuation* of early-cycle state-primary signal over 2008–2024 (H3). (The results show that the residual state-primary timing component is indistinguishable from zero throughout.) One plausible account, which is not directly tested in this paper, is infrastructural. English Wikipedia is organized by topics rather than jurisdictions, and its coordination devices, ranging from watchlists and WikiProjects to cross-page monitoring routines (Forte et al. 2014; Konieczny 2010), can pool attention and labor across states, eventually making candidate pages part of a shared national work queue rather than locally timed artifacts. Under that account, nationally synchronized production could be an endogenous property of the platform’s governance and tooling as much as a reflection of external political nationalization, especially in this incumbent-heavy subset of pages that attract routine

maintenance.

Identical editing rhythms can also be generated by alternative *informational* processes. Pages may be revised in response to nationally salient events without reproducing nationalized frames in their content, while locally meaningful developments may produce only modest aggregate editing volume. Accordingly, it would require direct analysis of edit content to convincingly claim that the content of candidate information has itself been nationalized. Future work could thus connect temporal patterns to the semantic and organizational character of edits to specify the mechanisms that remain opaque here.

7.3. *Revisiting the Three Mechanisms*

The evidence speaks most directly to downplaying the third mechanism, *strategic information production*, and specifically to its geographic organization. If interested actors were systematically timing their editing of candidate pages to coincide with local primaries, this would appear as a positive state-primary timing signal, especially in the weeks immediately before the primary when the informational stakes are highest. The flat Q2 profile argues against geographically targeted strategic editing as a *systematic, average* phenomenon across the candidates in this sample. This does not rule out strategic editing in particular high-profile races. Indeed, prior work documents exactly such activity on pages of sitting U.S. senators (Kalla & Aronow 2015) and German Bundestag members (Göbel & Munzert 2017), but it does suggest that across the broader population of House candidates with Wikipedia pages, any such activity is either rare enough or effectively enough countered by Wikipedia’s

quality-control mechanisms (like reversions) that it does not register in the aggregate timing profile.

The editor portfolio analysis lends further support to this interpretation. The editors who dominate editing volume on candidate pages are cross-party generalists with expansive geographic footprints. Such a profile is inconsistent with narrowly interested actors managing their preferred candidates' pages. This pattern is consistent with the "power of the few" structure documented throughout the Wikipedia literature (Kittur et al. 2007), but it adds a politically specific dimension: on candidate pages, the "few" are likely not partisans but caretakers.

The first two mechanisms, *reactive supply* and *electoral engagement*, are harder to distinguish empirically but both appear to operate at a national level. The pre-election ramp is temporally consistent with editors responding to the increasing volume of nationally circulated campaign news (reactive supply) and with the general heightening of political attention as the election approaches (electoral engagement). The composition-shift analysis suggests that these mechanisms primarily intensify the activity of the existing editing workforce rather than drawing in a qualitatively different set of contributors. The stable cross-party share is the stronger evidence on this point; the modest increase in geographic concentration leaves open the possibility that editors, whether the same individuals or new entrants, tilt somewhat toward home-state candidates as the election approaches.

7.4. *Limitations and Scope Conditions*

Several limitations bear on the interpretation of these results. First, the sample is non-random in ways that could attenuate the estimated state-primary timing signal. The set of House candidates with established Wikipedia pages is heavily skewed toward incumbents, notable politicians, and general-election winners, who are precisely the candidates whose pages are likely to be maintained year-round by the generalist editors documented in the EDA. Candidates *without* Wikipedia pages, or whose pages are created mid-cycle, are excluded by design. These excluded candidates are arguably the ones whose pages would be most likely to exhibit sharply primary-timed, episodic editing if such editing occurs. The Q2 null therefore characterizes the subset of candidate pages with sustained editorial attention, and should not be generalized to all candidates in House races.

Second, the editing process is sparse and bursty at the individual page level. Weekly aggregation and the separation of intensive and extensive margins mitigate this, but mean profiles can still dampen episodic spikes that occur in a small subset of races or cycles. The power analysis in Section 5 provides some reassurance that effects of policy-relevant magnitude would have been detectable, but the possibility remains that state-primary timing effects exist in specific subgroups (e.g., open-seat races with competitive primaries) and are averaged out in the panel estimates.

Third, as discussed in 7.2, this paper documents patterns in *information production*, not in the content or quality of information produced, and not in electoral behavior or voter decision-making. A candidate page that receives ten edits in a week

could be substantively improved or could be the site of a revert war that leaves it worse than before. Content-level analysis using machine learning or language models would be necessary to connect production dynamics to information quality (Rupprechter et al. 2025).

In addition, the patterns documented here could partially reflect institutional changes internal to Wikipedia in addition to changes in the broader political information environment. Such changes include the maturation of election-related WikiProjects, evolving editorial policies, and the proliferation of semi-automated editing tools that reduce the friction of cross-state work. Due to the design constraint, this paper is unable to empirically distinguish these internal and external explanations, though the consistency of the Q2 null across all nine cycles (including the earlier period before many such tools were widely adopted) suggests that platform-internal evolution is not the sole driver.

A further consideration is that Wikipedia’s page protection mechanism may shape the editing patterns observed in this study. Candidate pages that attract contentious or vandalistic editing, which likely increases as elections approach and pages become more visible can be semi-protected (restricting edits to registered accounts) or fully protected (restricting edits to administrators) by Wikipedia administrators (Merkley 2020). If protections are disproportionately applied during the pre-election window, the observed editing ramp in Q1 may understate the latent demand for editing by filtering out contributions that would otherwise have occurred. Moreover, protections act as a compositional filter, since semi-protection mechanically excludes IP editors, which could partly explain both the smaller pre-election ramp among IP ed-

itors (Appendix D) and the dominance of registered accounts in the near-election editing workforce. To the extent that the editors filtered out by protection differ systematically from those who remain (for instance, if locally motivated or partisan editors are more likely to be unregistered) the composition-shift analysis may present an artificially stable picture of the editing workforce across time windows. Page protection is thus both a feature of Wikipedia’s quality-control infrastructure and a potential source of selection in the data. Disentangling its effects from the underlying production dynamics would require page-level protection histories that this study did not investigate due to time constraints.

8. Conclusion

This paper has examined the temporal and organizational structure of Wikipedia editing on U.S. House candidate biography pages across nine election cycles (2008–2024). The analysis yields three principal findings. First, candidate-page editing displays a modest pre-election ramp that is nationally synchronized, with the dominant production event being a large post-election updating burst consistent with Wikipedia’s institutional commitment to documenting verified facts (Merkley 2020). Second, the state-primary timing component—the residual editing variation attributable to proximity to a state’s own primary, net of national synchronization—is indistinguishable from zero across all cycles, window choices, and trend specifications tested. Third, the editing workforce is increasingly geographically expansive, with the highest-volume contributors tending to edit candidates of both parties across many states in a pattern more consistent with informational maintenance than with partisan advocacy.

A comparison of editing composition between baseline and near-general-election windows finds no shift in partisan composition, though editors in the pre-election window are slightly more geographically concentrated in their top state—a modest tilt that does not fundamentally alter the cross-state character of the workforce.

These findings contribute to the American politics literature by documenting that the production side of a major reference platform mirrors the consumption-side patterns already established in works on Wikipedia pageview and forecasting (Yasseri & Bright 2016; Smith & Gustafson 2017). The supply of political information on Wikipedia, like voter attention and campaign spending, appears to be organized around a national electoral calendar rather than around the state-specific institutional calendars that formally structure American primary elections. Whether this reflects the broader nationalization of American politics (Hopkins 2018; Moskowitz 2021) or the inherently national architecture of Wikipedia as a platform, or some combination of both, the practical consequence is the same: the information that voters and increasingly large language models encounter on Wikipedia candidate pages is produced by a nationally integrated, predominantly nonpartisan editing workforce operating on a national timeline.

Future research following directly from the limitations of this study can use off-ballot candidate pages as diff-in-diff controls to allow cleaner attribution of the pre-election ramp to electoral factors as opposed to seasonal variation, employ language-inference-based content analysis to further characterize the quality of information available to voters, and extend the analysis to U.S. Senate and state legislative races. Moreover, given how Wikipedia content is increasingly used as training data for large

language models, the way biases and production patterns of candidate information propagate through downstream systems merits separate investigation.

Notes

¹except Louisiana, which is dropped from the data.

²The election-year clustering dimension has only 9 clusters, which can make CRV standard errors conservative or anticonservative depending on the data structure. Results are qualitatively the same when clustering by page-cycle only; both are reported in Appendix B.

³The extensive margin is omitted because the identification strategy in Q2 relies on within-calendar-week variation in distance to primary; the binary P_{jt} outcome would compress this variation and yield very low statistical power once calendar-week fixed effects absorb the nationally shared activation pattern.

⁴Because state primaries occur at different calendar dates, the Q2 design is structurally a staggered event study, which raises the question of whether the TWFE estimator suffers from the negative-weighting problems identified by Goodman-Bacon (2021) and Sun and Abraham (2021). However, this concern should be mitigated by the fact that the “treatment” here is transient rather than absorbing. Since each page-cycle passes through the event-time window $k \in \{-18, \dots, -1\}$ and then exits, already-treated units do not persistently contaminate the control group for later-treated units.

⁵The sequence skips $Trend = 6$ because 2020, which would correspond to $(2020 - 2008)/2 = 6$, is excluded.

⁶The minimum detectable effect is $(z_{\alpha/2} + z_{\beta}) \times SE = (1.96 + 0.84) \times 0.0132 \approx 0.037$, where $z_{\alpha/2} = 1.96$ is the two-sided critical value at $\alpha = 0.05$ and $z_{\beta} = 0.84$ corresponds to 80% power.

⁷any “guest or unregistered users who edited Wikipedia without being logged into an account” (Wikipedia:IP users 2026).

⁸State footprint $S_{e,y}$ is mechanically inflated in the baseline window (which spans roughly 50 weeks) relative to the 18-week near-general window, because editors have more time to accumulate edits across states. MS avoids this confound.

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A. Technical Design Details

This appendix summarizes technical details of the panel construction and regression specifications.

Panel construction. The unit of analysis is a candidate-page $\times$ two-year election cycle $\times$ week, where weeks are defined as Tuesday–Monday intervals and cycles span the two years preceding each general election from 2008 to 2024. The post-election updating period (from the prior general election through the week Congress convenes) is dropped to isolate pre-election dynamics, and pages created mid-cycle are excluded.

Q1 outcome variables. For each page-cycle j and week t , baseline-centering is not used for Q1 (see Section 3).

Q2 baseline-centering. For Q2, editing is centered on each page-cycle’s mean over weeks before the 18-week near-primary window ($k^{(P)} \leq -19$): $\tilde{Y}_{j,t}^{(P)} = Y_{j,t} - \bar{Y}_j^{(P)}$. This centering removes stable level differences before calendar-week FEs absorb remaining national synchronization.

Q2 model 1 (Equation 2). $\tilde{Y}_{j,t}^{(P)} = \alpha_j + \delta_t + \sum_{k=-18}^{-1} \beta_k D_k^P(j, t) + \varepsilon_{j,t}$. FEs: page-cycle α_j , calendar-week δ_t . SE: two-way clustered by page-cycle and calendar-week.

Q2 model 2 (Equation 3). $\tilde{Y}_{j,t}^{(P)} = \alpha_j + \delta_t + \sum_{k=-18}^{-1} (\beta_k + \gamma_k \cdot Trend_y) D_k^P(j, t) + \varepsilon_{j,t}$, where $Trend_y = (y - 2008)/2 \in \{0, 1, 2, 3, 4, 5, 7, 8\}$ (eight cycles; 2020 excluded). H2 tested via $\bar{\beta}_K^{(M1)} = \frac{1}{|K|} \sum_{k \in K} \hat{\beta}_k$ from Model 1; H3 tested via $\bar{\gamma}_K = \frac{1}{|K|} \sum_{k \in K} \gamma_k$ from

Model 2. Both reported across $K \in \{\{-4, -3, -2, -1\}, \{-2, -1\}, \{-8, \dots, -1\}\}$.

Year-by-year $PTI_y = \frac{1}{4} \sum_{k=-4}^{-1} f(k, y)$ from Model 2 appears in Appendix Table 5.

B. Q1 Clustering Robustness

Table 4 compares the Q1 National Index estimates under two variance specifications: the baseline two-way clustering (page-cycle $\times$ election-year) used in Table 1, and one-way clustering by page-cycle only. Because the election-year dimension has only nine clusters, the two-way clustered standard errors are substantially larger. Point estimates are identical. The qualitative conclusion, a positive and statistically distinguishable pre-election editing buildup, is robust to the choice of clustering.

Table 4. Q1 National Index: Two-Way vs. One-Way Clustered Standard Errors.

Outcome	Window	NI	$SE_{2\text{-way}}$	p2-way	$SE_{1\text{-way}}$	p1-way
Intensive (log edits)	$k \in \{-8, \dots, -1\}$	0.0911	0.0347	0.009***	0.0054	<0.001***
Intensive (log edits)	$k \in \{-10, \dots, -1\}$	0.0802	0.0305	0.009***	0.0051	<0.001***
Extensive ($P > 0$)	$k \in \{-8, \dots, -1\}$	0.0695	0.0286	0.015**	0.0036	<0.001***
Extensive ($P > 0$)	$k \in \{-10, \dots, -1\}$	0.0629	0.0263	0.017**	0.0033	<0.001***

Note: Both columns report the same point estimates from the Q1 event-study regression (Equation 1); only the variance estimator differs. Two-way: CRV1 clustered by page-cycle and election-year (9 year clusters). One-way: CRV1 clustered by page-cycle only (3,739 clusters). *** $p < 0.01$; ** $p < 0.05$.

C. Registered and IP Editors Side-by-Side

The following figures show the original side-by-side EDA plots for registered and IP editors. IP editors exhibit substantially higher variance, because only a small effective sample passes the activity threshold ($N \geq 5$ edits) and low-activity, one-off IP contributors dominate.

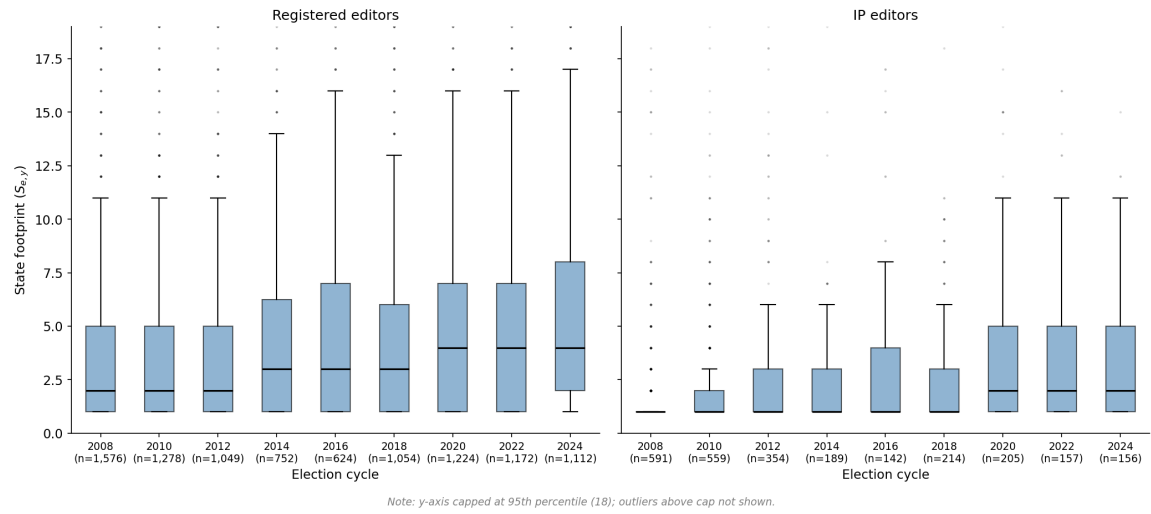


Figure 12. State Footprint by Election Cycle: Registered vs. IP Editors.
Note: Distribution of $S_{e,y}$ for registered (left) and IP (right) editors with $N_{e,y} \geq 5$.

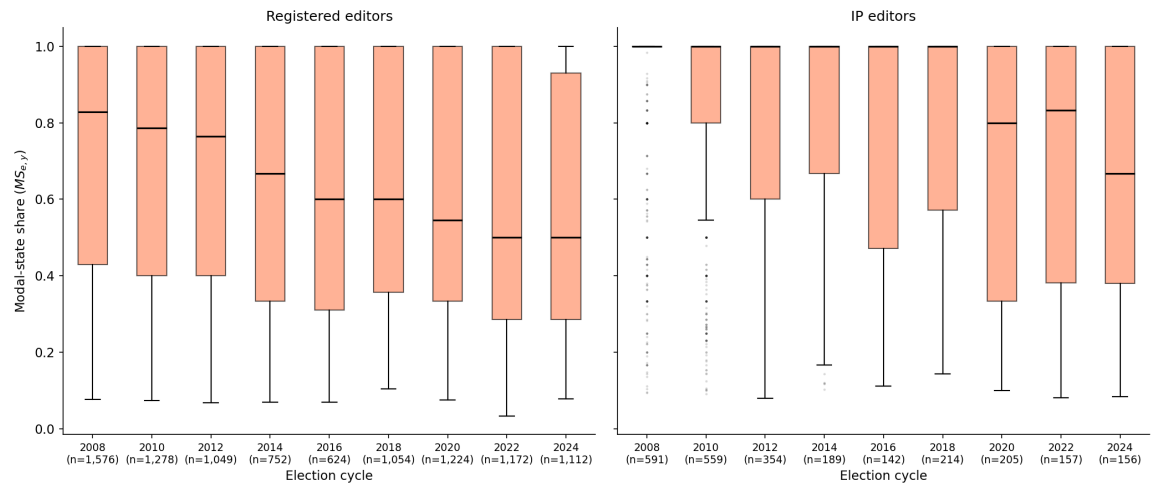


Figure 13. Modal-State Share by Election Cycle: Registered vs. IP Editors.
Note: Distribution of $MS_{e,y}$ for registered (left) and IP (right) editors with $N_{e,y} \geq 5$.

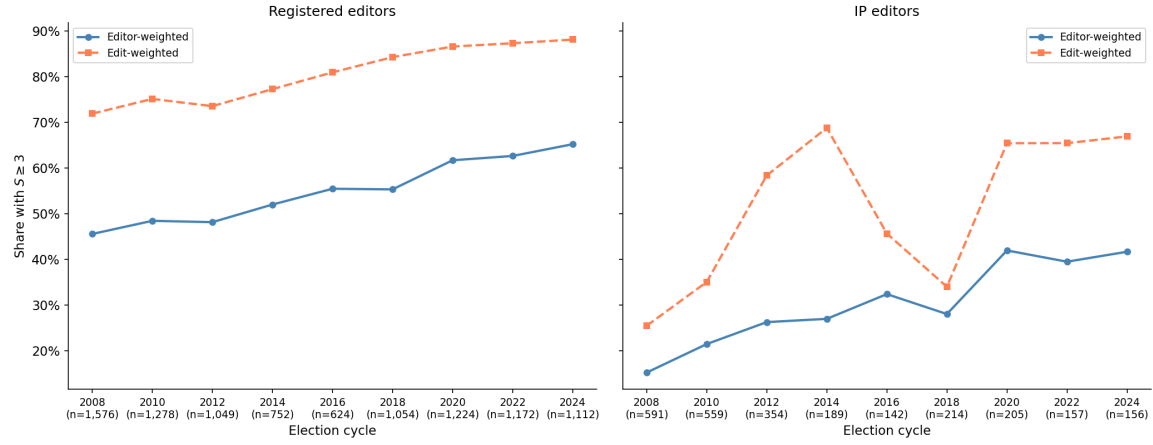


Figure 14. Multi-State Share over Cycles: Registered vs. IP Editors.

Note: Fraction of editors (editor-weighted) or edits (edit-weighted) from editors with $S_{e,y} \geq 3$, for registered (left) and IP (right) editors with $N_{e,y} \geq 5$.

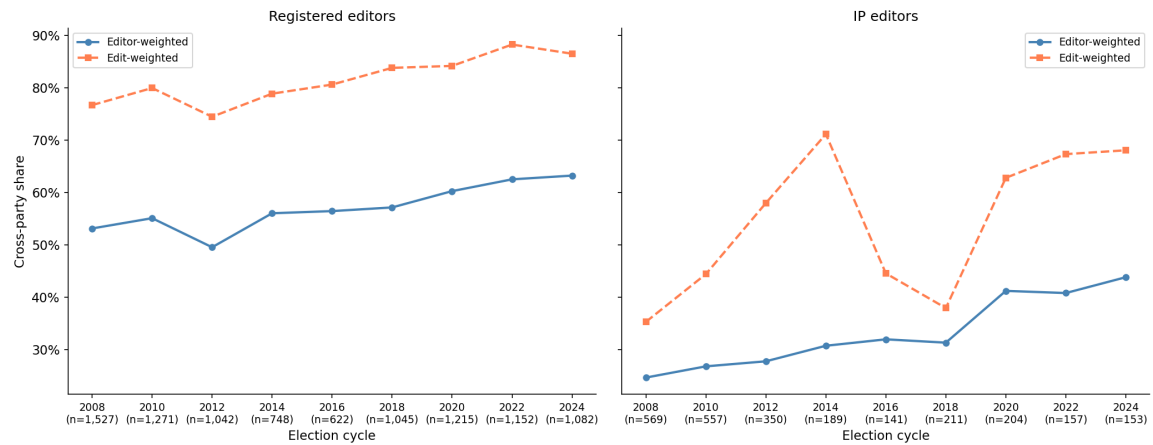


Figure 15. Cross-Party Participation by Cycle: Registered vs. IP Editors.

Note: Fraction of editors (editor-weighted) or edits (edit-weighted) from editors with $N_{e,y}^D > 0$ and $N_{e,y}^R > 0$, for registered (left) and IP (right) editors with $N_{e,y}^D + N_{e,y}^R \geq 5$.

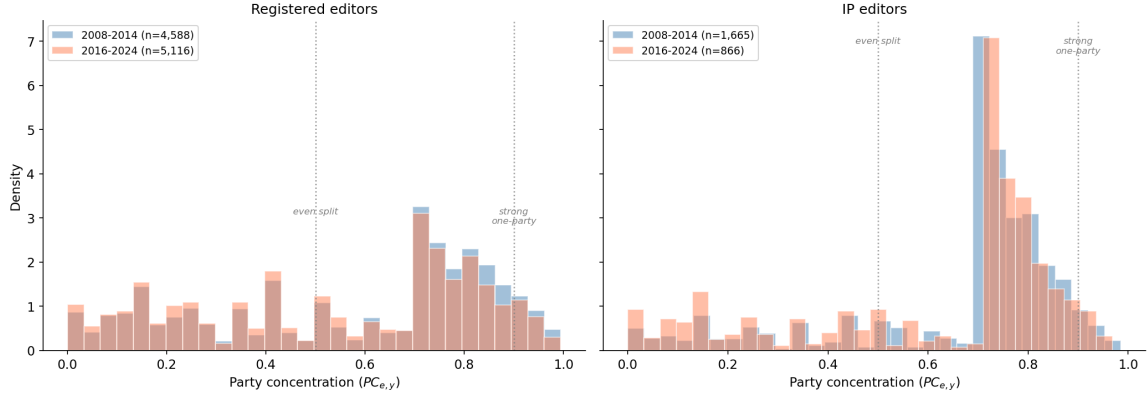


Figure 16. Party Concentration Index: Registered vs. IP Editors.

Note: Distribution of $\pi_{e,y} = (N_{e,y}^D + 1)/(N_{e,y}^D + N_{e,y}^R + 2)$ pooled into early (2008–2016) and late (2018–2024) periods, for registered (left) and IP (right) editors with $N_{e,y}^D + N_{e,y}^R \geq 5$.

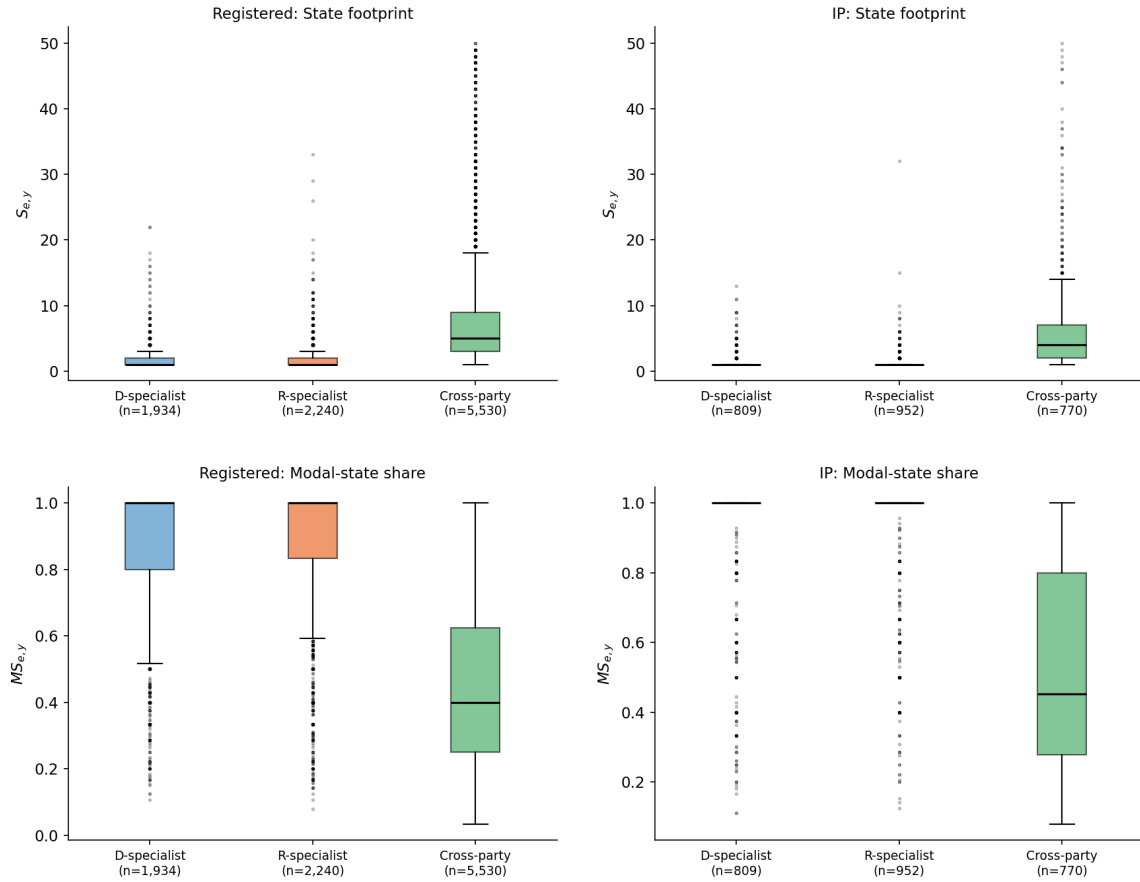


Figure 17. Geographic Scope by Party Type: Registered vs. IP Editors.

Note: State footprint $S_{e,y}$ and modal-state share $MS_{e,y}$ by party type (cross-party, D-specialist, R-specialist), pooled across cycles, for registered (left) and IP (right) editors with $N_{e,y}^D + N_{e,y}^R \geq 5$.

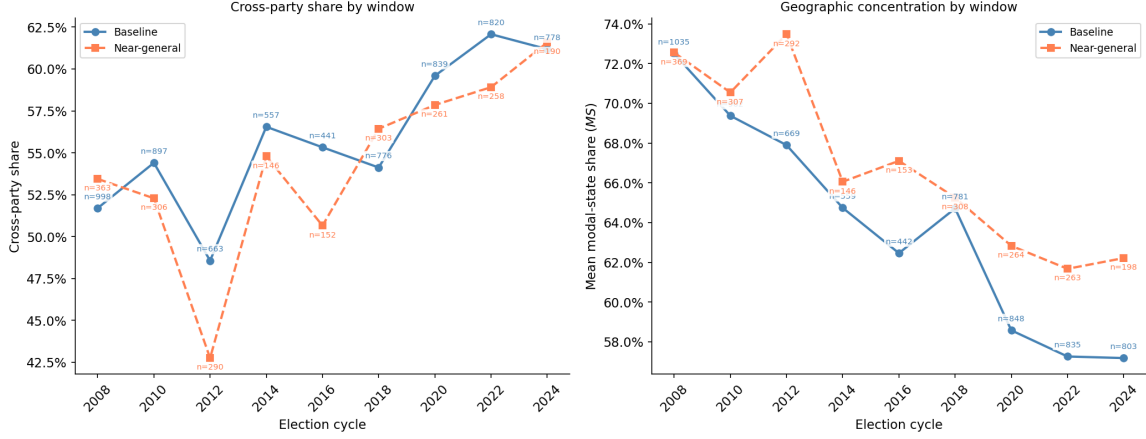


Figure 18. Composition Shift (Baseline vs. Near-General Window) for Registered Editors.

Note: Same as Figure 11. Shown here for comparison with Figure 12. Cross-party participation share (left) and mean modal-state share MS (right) in the baseline vs. near-general-election window ($k^{(G)} \in \{-18, \dots, -1\}$), by cycle. Registered editors with $N_{e,y}^D + N_{e,y}^R \geq 5$ (cross-party) or $N_{e,y} \geq 5$ (geographic).

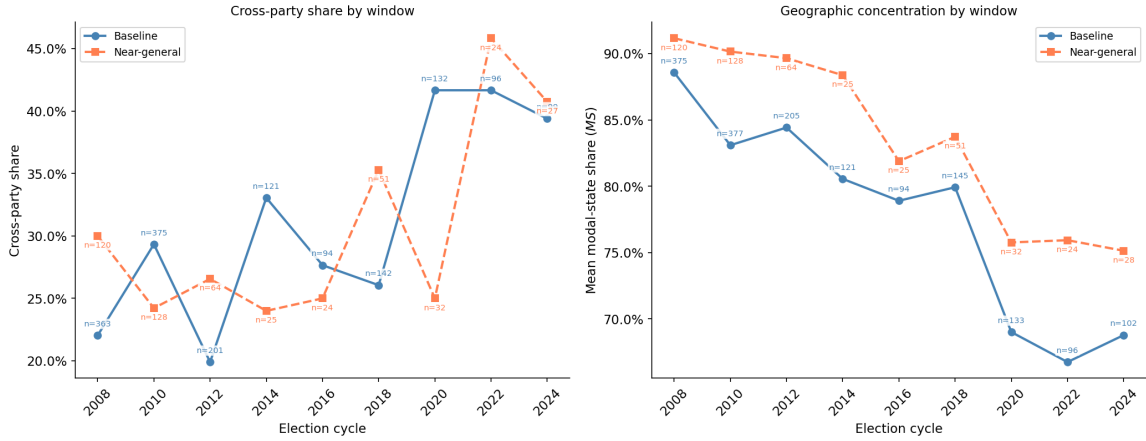


Figure 19. Composition Shift (Baseline vs. Near-General Window) for IP Editors.

Note: Cross-party participation share (left) and mean modal-state share MS (right) in the baseline vs. near-general-election window ($k^{(G)} \in \{-18, \dots, -1\}$), by cycle. IP editors with $N_{e,y}^D + N_{e,y}^R \geq 5$ (cross-party) or $N_{e,y} \geq 5$ (geographic).

D. Q1: IP (Anonymous) Editing over the Election Cycle

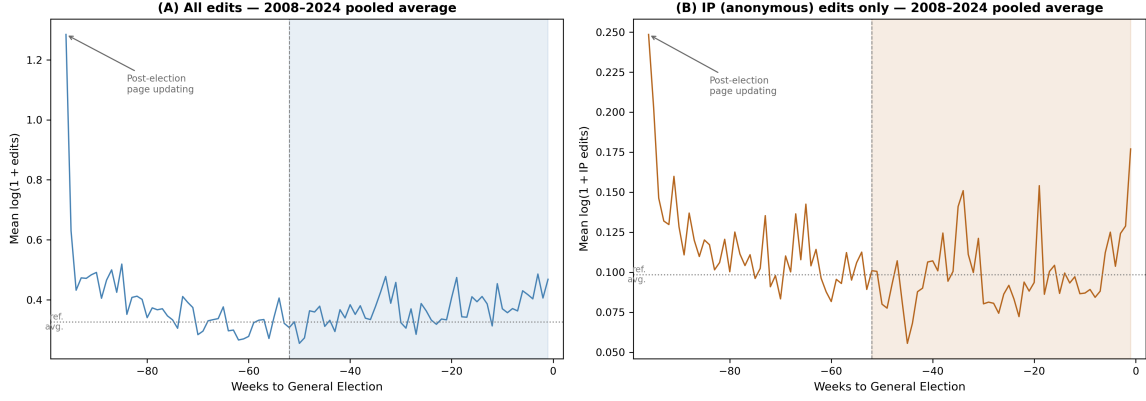


Figure 20. Election-Cycle Editing Trajectories: All Edits vs. IP Edits Only.

Note: Panel A reproduces the all-edits trajectory from Figure 1; Panel B shows IP (anonymous) edits only, 2008–2024 pooled average. y -axis: $\log(1 + \text{edits})$ averaged across all page-cycle-weeks at each k . Horizontal dotted line: reference-period average ($k \in \{-70, \dots, -36\}$). In the pre-election window $k \in \{-8, \dots, -1\}$, all edits average 28% above the reference level; IP edits average 20% above their own reference level. The post-election updating burst is proportionally larger for registered editors, suggesting that post-results factual updating is concentrated among returning account holders.

E. Q2 Year-by-Year Primary-Timing Index

Table 5. Year-by-Year Primary-Timing Index from Model 2.

Year	PTI_y	SE	Wald	p-value
2008	0.0020	0.0249	0.0067	0.9350
2010	0.0003	0.0206	0.0002	0.9895
2012	-0.0015	0.0168	0.0078	0.9295
2014	-0.0032	0.0141	0.0533	0.8175
2016	-0.0050	0.0130	0.1492	0.6993
2018	-0.0068	0.0139	0.2370	0.6264
2022	-0.0103	0.0202	0.2585	0.6112
2024	-0.0120	0.0245	0.2418	0.6229

Note: $PTI_y = \frac{1}{4} \sum_{k=-4}^{-1} f(k, y)$ from Model 2, main window $K = \{-4, -3, -2, -1\}$. These are arithmetic projections from the scalar parameters $\bar{\beta}_K^{(M1)}$ and $\bar{\gamma}_K$ in Tables 2 and 3; they confirm that no individual cycle shows a statistically significant state-primary timing effect. The 2020 cycle is excluded due to COVID-19 disruptions to state primary scheduling. $H_0: PTI_y = 0$, tested via 1-df Wald test; SE: two-way clustered (page-cycle $\times$ calendar-week).